

Chapter 1

Electrochemistry and Corrosion Science

1.1 Electrochemistry

Electrochemistry deals with the chemical reactions produced by passing electric current through an electrolyte or the production of electric current through chemical reactions. The cells which convert chemical energy to electric energy are called electro chemical cells and the cells which convert electric energy to chemical energy are called electrolytic cells. Differences between electrochemical cells and electrolytic cells are summarized in table 1.1.

Table 1.1: Differences between electrochemical cells and electrolytic cells

Electrochemical cells	Electrolytic cells
Convert chemical energy to electric energy	Convert electric energy to chemical energy
Cell emf is +ve	Cell emf is -ve
Free energy change ΔG is -ve	Free energy change ΔG is +ve
Anode is -ve terminal	Anode is +ve terminal

1.1.1 Electrochemical cell

An electrochemical cell (voltaic or galvanic cell) is a device in which the free energy of a physical or chemical process is converted into electrical energy. Such a cell usually consists of two electrodes immersed in one or more suitable electrolytes. Eg. Daniel cell.

Daniel cell consists of a Zn electrode immersed in $ZnSO_4$ solution in a beaker and a Cu electrode immersed in $CuSO_4$ solution in a beaker and they are connected by means of salt bridge or porous partition.

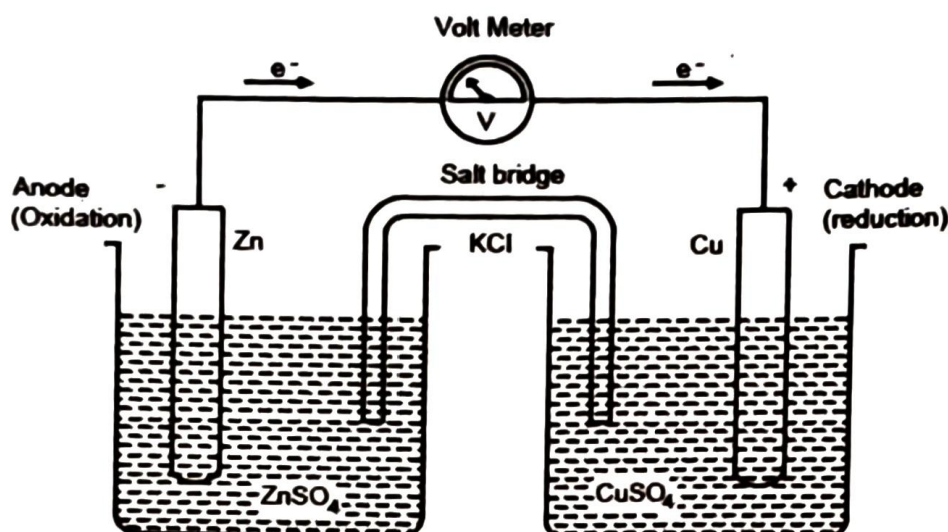


Figure 1.1: A schematic view of a Daniel cell

Anode (-) Zinc metal has greater tendency for oxidation than that of copper, hence oxidation occurs in the Zn metal and Zn atom leaves its electrons to the Zn electrode and goes to the solution as Zn^{2+} , hence, the Zn electrode is the anode and acquires a negative charge.



Cathode (+) In the Cu compartment, Cu^{2+} in the solution has a tendency of reduction, it accepts electrons from the Cu metal and get deposited on Cu electrode.



The electrons generated at the Zn electrode are consumed at the copper electrode. The flow of electrons (negative electricity) in the cell is from the Zn electrode to the Cu electrode. Consequently, the flow of positive electricity i.e. the conventional current in the cell is from Cu electrode to the Zn electrode. The cell reaction can be obtained by adding equation (1.1) and (1.2)



When the electrons flow through the external wire, for the completion of circuit, there should be the flow of ions between the compartments. For the transference of ions between the compartments a salt bridge or a porous partition is used. The Daniel cell may be represented as



Anode is written on the left and cathode is written on the right, the '//' represents the salt bridge, metal and its ions are separated by '/' and the ionic species are written on either side of the salt bridge.

Note: Anode is the place where oxidation takes place, its sign is negative in electrochemical cell and positive in electrolytic cells. For Cathode it is the reverse.

Functions of Salt Bridge

1. Provides the continuity of electrical circuit. When electrons flow through the external wire, there should be a flow of ions between the compartments.
2. It prevents the diffusion of interfering ions from one compartment to another. For example, in Daniel cell the diffusion of Cu^{2+} to the Zinc compartment has to be prevented as it gets deposited over the Zinc metal.
3. Salts like KCl , NH_4NO_3 are used for making salt bridge. Ions like K^+ and Cl^- has equal mobility and get drifted to cathode and anode compartments respectively under the potential difference.

1.1.2 Reversible and Irreversible Cells

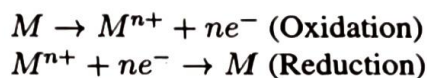
Daniel cell has the emf value 1.1 volt. If an opposing emf exactly equal to 1.1 volt is applied to the cell, the cell reaction, $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$ stops but if it is increased infinitesimally beyond 1.1 volt, the cell reaction is reversed i.e; $Zn^{2+} + Cu \rightarrow Zn + Cu^{2+}$. Such a cell is termed as a reversible cell. Thus, the following are the two main conditions of reversibility.

1. The chemical reaction of the cell stops when an exactly equal and opposing emf is applied.
2. The chemical reaction of the cell is reversed and the current flows in opposite direction when the opposing emf is slightly greater than that of the cell.

Any other cell which does not obey the above two conditions is termed as irreversible. The first electrochemical cell was developed by Alexander Volta and is called Volta cell. It is an example for an irreversible cell. In Volta cell, a Zinc rod and a Copper rod are dipped in a solution of dilute sulphuric acid. Zinc acts as anode and oxidation of zinc takes place and Cu acts as cathode and reduction of H^+ to H_2 gas takes place here. On applying the reverse potential greater than cell emf, oxidation of copper takes place, since no H_2 gas is available for oxidation. Reversibility does not mean rechargeability, Daniel cell cannot be recharged. When charging current is more H_2 evolution takes place at Zn electrode rather than the reduction of Zn^{2+} .

1.1.3 Single Electrode Potential and Helmholtz electrical double layer

When a metal electrode is in contact with its own salt solution ($M^{n+}nX^-$) the following reaction can occur due to the tendency called Nernst electrolytic solution tension.



In the first case some metal ions enter into the electrolyte solution leaving behind the electrons on the electrode. Thus, electrode gets negative charge and solution side gets positive charge due to excess positive metal ions. In the second case the positive metallic ions from the solution take electrons from the electrode and get deposited as metal on the electrode surface. Now the electrode gets a positive charge and the solution side gets a negative charge (due to deficiency of positive metal ions).

Equilibrium is reached in the vicinity of the electrode, due to the electrostatic force; any further transference of metal ions does not take place. Thus, the positive charge and negative charge remain close to the metal surface forming a double layer. This is called Helmholtz electrical double layer. As a result, a difference in potential is set up between the metal and the solution, this potential is called single electrode potential.

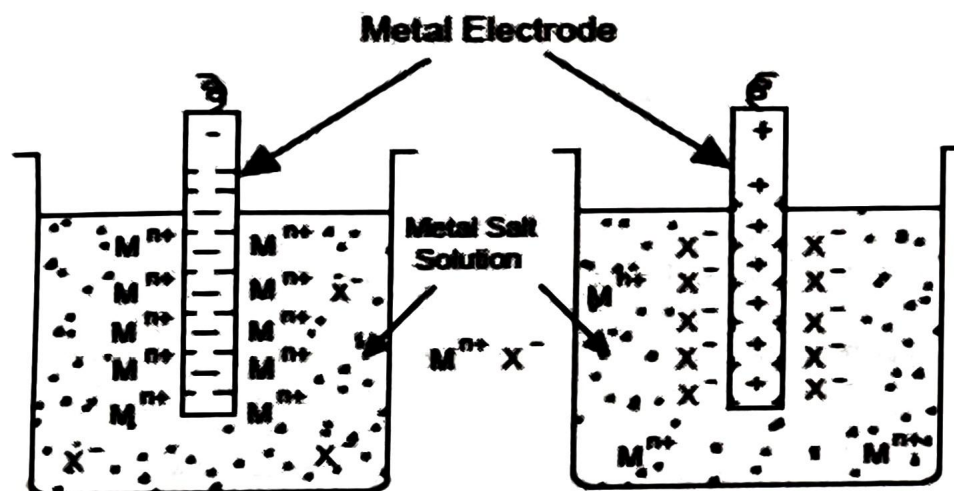


Figure 1.2: Helmholtz electrical double layer

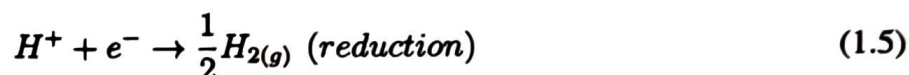
The tendency of an electrode to lose or gain electrons when in contact with its own ionic solution is termed as electrode potential. The tendency of the electrode to lose electrons is called oxidation potential and the tendency of the electrode to gain electrons is called reduction potential. This potential difference becomes constant at equilibrium. This constant potential difference developed when a metal is in contact with its own salt solution of concentration 1 M at 25°C is called the standard electrode potential of the metal.

It is impossible to experimentally determine the potential of a single electrode. Only the difference in potentials between the two electrodes can be measured. It is possible to find out the potential of single electrode by arbitrarily fixing the potential of one of the electrodes as zero. Hence, the potential of standard hydrogen electrode (SHE) has been fixed as zero, which is known as the primary reference electrode.

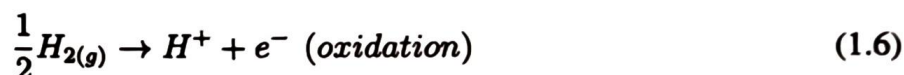
1.1.4 Standard Hydrogen Electrode (SHE) or Normal Hydrogen Electrode (NHE)

SHE is an example of gas electrode. It consists of an inert platinum electrode dipped in 1M H^+ (HCl) solution, which is in contact with H_2 gas at 1 atm pressure at 298 K. SHE is taken as the standard reference electrode and the potential of which is taken as zero. The electrode can be represented as $Pt/H_{2(g)1atm}/H^+_{(1M)}$.

SHE is a reversible electrode which can act as cathode or anode depending on the potential of the electrode to which it is coupled. If the potential of the coupled electrode is less than zero (eg. Zn, Mg, Li etc), reduction reaction takes place in SHE and it act as cathode as shown below



When the potential of the coupled electrode is greater than zero eg. Cu, Ag, F_2 , oxidation reaction takes place in SHE and it acts as anode, as shown



Standard hydrogen electrode is the primary reference electrode used for finding single electrode potential of other electrodes. The -ve or +ve values obtained for standard reduction/electrode potential depends on whether the half-cell acts as anode or cathode with respect to SHE. For example, standard electrode potential of zinc electrode is -ve since the electrode acts as anode with respect to SHE.

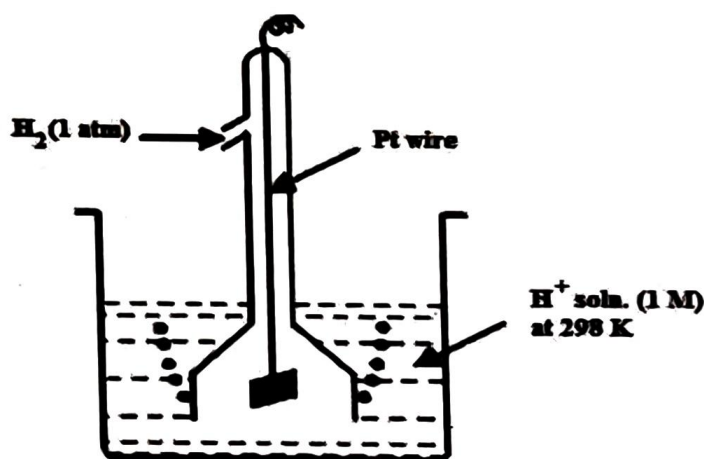


Figure 1.3: Standard Hydrogen electrode

Limitations of SHE

1. It is very cumbersome to set up a SHE.

2. It requires considerable volume of solution.
3. The hydrogen electrode will be poisoned by species like Hg, As, S, Fe^{3+} , MnO_4^- etc.
4. It cannot be used in solutions containing redox systems.
5. Difficult to maintain pressure of H_2 gas at 1 atm and difficult to maintain the concentration of 1 M HCl.

1.1.5 Electrochemical series

It is the series in which the standard reduction potentials are arranged in descending order (table 1.2). According to IUPAC system only reduction reactions and reduction potentials are taken into consideration.

Table 1.2: Standard reduction potentials at 298 K in electrochemical order

Reduction half Reaction	E^0 in V
$F_2 + 2e \rightarrow 2F^-$	+ 2.87
$S_2O_8^{2-} + 2e \rightarrow 2SO_4^{2-}$	+ 1.98
$Ce^{4+} + e \rightarrow Ce^{3+}$	+ 1.67
$MnO_4^- + 8H^+ + 5e \rightarrow Mn^{2+} + 4H_2O$	+ 1.51
$Au^{3+} + 3e \rightarrow Au$	+ 1.40
$Cl_2 + 2e \rightarrow 2Cl^-$	+ 1.36
$Cr_2O_7^{2-} + 14H^+ + 6e \rightarrow 2Cr^{3+} + 7H_2O$	+ 1.33
$O_2 + 4H^+ + 4e \rightarrow 2H_2O$ (1M H^+ acid medium)	+ 1.23
$NO_3^- + 2H^+ + e \rightarrow NO_2 + H_2O$	+ 0.80
$Ag^+ + e \rightarrow Ag$	+ 0.80
$Fe^{3+} + e \rightarrow Fe^{2+}$	+ 0.77
$I_2 + 2e \rightarrow 2I^-$	+ 0.54
$O_2 + 2H_2O + 4e \rightarrow 4OH^-$ (1M OH^- base medium)	+ 0.40
$Cu^{2+} + 2e \rightarrow Cu$	+ 0.34
$Hg_2Cl_2 + 2e \rightarrow 2Hg + 2Cl^-$	+ 0.28
$AgCl + e \rightarrow Ag + Cl^-$	+ 0.22
$2H^+ + 2e \rightarrow H_2$ (1M H^+ acid medium)	0
$Fe^{3+} + 3e \rightarrow Fe$	0.04
$Ni^{2+} + 2e \rightarrow Ni$	- 0.26
$Fe^{2+} + 2e \rightarrow Fe$	- 0.44
$Zn^{2+} + 2e \rightarrow Zn$	- 0.76
$2H_2O + 2e \rightarrow H_2 + 2OH^-$ (1M OH^- base medium)	- 0.83
$Al^{3+} + 3e \rightarrow Al$	- 1.66
$Mg^{2+} + 2e \rightarrow Mg$	- 2.36
$Na^+ + e \rightarrow Na$	- 2.71
$K^+ + e \rightarrow K$	- 2.93
$Li^+ + e \rightarrow Li$	- 3.05

1.1.5.1 Applications of electrochemical series

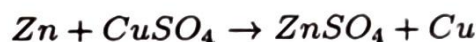
1. Reactivity of metals

Electrodes having negative values of standard electrode potential with respect to hydrogen have the tendency to undergo oxidation whereas electrodes having positive values have the tendency to undergo reduction in aqueous solution.

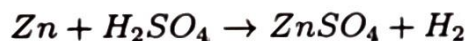
For example, Na metal has high negative value of reduction potential and possesses tendency for oxidation to Na^+ ion, whereas Au^{3+} has positive value of reduction potential, hence possesses tendency for reduction to Au metal.

2. Displacement reactions

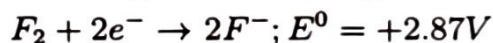
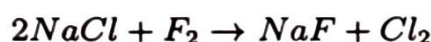
- (a) **To predict whether a given metal will displace another, from its salt solution:** A metal having lower E^0 value will displace the metal having higher E^0 value from its salt solution. Eg: Zn metal can displace Cu from $CuSO_4$ solution.



- (b) **Metals having negative reduction potentials can displace hydrogen gas from acids.** For example, metals like Fe, Zn, Al etc. can liberate H_2 gas when react with acids.



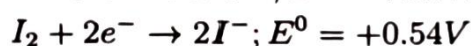
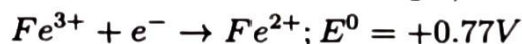
- (c) **A non-metal having higher reduction potential can displace a non-metal having lower reduction potential.** For example, if we pass fluorine gas through sodium chloride solution, displacement reaction takes place generating chlorine gas and sodium fluoride



3. **Determination of standard emf of a cell:** Standard emf of a cell can be calculated from the standard electrode potentials of respective electrodes.

$$E_{cell}^0 = E_{cathode}^0 - E_{anode}^0 \quad (1.7)$$

4. **Spontaneity of redox reactions:** The spontaneity of a chemical reaction can be judged based on the standard reduction potential. For example, if we add potassium dichromate solution to an acidified solution of mixture of ferrous chloride and potassium iodide, the oxidation of I^- to I_2 takes place first than oxidation of Fe^{2+} to Fe^{3+} . The reaction having more cell emf is more spontaneous which is evident from following reduction potentials.



Similarly acidified dichromate solution is not capable of liberating Cl_2 gas from $NaCl$, but acidified permanganate can liberate Cl_2 gas from $NaCl$, which can be explained from following reduction potentials.

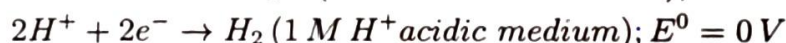


The cell emf of permanganate- Cl_2/Cl^- cell is positive and the reaction is spontaneous, whereas cell emf of dichromate- Cl_2/Cl^- cell is -ve and the reaction is non spontaneous.

5. Analysis of products of electrolysis:

If we electrolyze a mixture of aqueous solution of $NaCl$ and NaI using platinum electrodes, H_2 is liberated at cathode (-ve terminal) and I_2 is liberated at +ve terminal. The ions approaching -ve terminal are H^+ and Na^+ . The species having highest reduction potential (less -ve) undergo reduction ($E_{H^+/H_2}^0 = -0.41$ V at pH = 7 and $E_{Na^+/Na}^0 = -2.71$ V). Thus, H_2 is liberated rather than Na metal. Similarly, the ions approaching +ve terminal are Cl^- , I^- and OH^- . The species having lowest reduction potential get oxidized ($E_{Cl_2/Cl^-}^0 = +1.36$ V, $E_{I_2/I^-}^0 = +0.54$ V and $E_{O_2/OH^-}^0 = +0.82$ V at pH = 7). Thus, I_2 is liberated than O_2 or Cl_2 gas.

6. **Decomposition potential:** Decomposition potential of water can be calculated from electro chemical series. Water can be split into hydrogen and oxygen just by applying 1.23 V which is evident from the following reduction potentials of oxygen and hydrogen electrodes.



1.1.6 Nernst equation for the electrode potential

It gives the effect of electrolyte concentration on electrode potential. Consider a general reduction reaction, taking place at the electrode, $M^{n+} + ne^- \rightarrow M_{(s)}$. The decrease in free energy, ΔG accompanying this process is given by the well-known thermodynamic equation. (Vant Hoff's equation)

$$\Delta G = \Delta G^0 + RT \ln Q \quad (1.8)$$

where Q is called reaction quotient, which is the ratio of activities of products to the reactant in a non-equilibrium situation (at equilibrium $\Delta G = 0$ and $Q = K$). In a reversible cell, electric energy is produced at the expense of decrease in free energy, and free energy is related with EMF of the cell as follows

$$\Delta G = -nFE \quad (1.9)$$

Substituting equation (1.9) in (1.8)

$$-nFE = -nFE^0 + RT \ln Q \quad (1.10)$$

Dividing by -nF

$$E = E^0 - \frac{RT}{nF} \ln Q \quad (1.11)$$

Substituting the values of constants at the temperature $T = 298\text{ K}$, $R = 8.314\text{ JK}^{-1}\text{mol}^{-1}$ and $F = 96500\text{ C}$ (Faraday).

$$E = E^0 - \frac{8.314 \times 298 \times 2.303}{n \times 96500} \log Q$$

$$E = E^0 - \frac{0.0591}{n} \log Q \quad (1.12)$$

Equations (1.11) and (1.12) are different forms of Nernst's equations.

Nernst Equation for single electrode: For a single electrode reaction $M_{aq}^{n+} + ne^- \rightarrow M(s)$ reaction quotient Q can be written as

$$Q = \frac{[M]}{[M^{n+}][e^-]^n}$$

In dilute solution, activities are replaced by molar concentration, also the concentration terms $[e^-]$ is taken as one and for pure solid $[M]$ is also taken as one, therefore the reaction quotient Q becomes

$$Q = \frac{1}{[M^{n+}]}$$

Substituting in equation (1.12)

$$E = E^0 - \frac{0.0591}{n} \log \frac{1}{[M^{n+}]} \quad (1.13)$$

Nernst Equation for a complete cell: For a cell reaction $aA + bB \rightarrow cC + dD$ reaction quotient Q can be written as

$$Q = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$

Modified form of equation (1.12) for complete cell is given by

$$E_{\text{cell}} = E_{\text{cell}}^0 - \frac{0.0591}{n} \log Q \quad (1.14)$$

$$E_{\text{cell}} = E_{\text{cell}}^0 - \frac{0.0591}{n} \log \frac{[C]^c[D]^d}{[A]^a[B]^b}$$

For example, the Daniel cell reaction $Zn_{(s)} + Cu_{(aq)}^{2+} \rightarrow Zn_{(aq)}^{2+} + Cu_{(s)}$

$$Q = \frac{[Zn_{(aq)}^{2+}][Cu_{(s)}]}{[Zn_{(s)}][Cu_{(aq)}^{2+}]}$$

Here concentrations of pure solids are taken as one, hence

$$Q = \frac{[Zn^{2+}]}{[Cu^{2+}]}$$

Substituting in equation (1.14)

$$E_{\text{cell}} = E_{\text{cell}}^0 - \frac{0.0591}{n} \log \frac{[Zn^{2+}]}{[Cu^{2+}]} \quad (1.15)$$

Solved Problems

Problem 1.1.1: Calculate the single electrode potential of Zn electrode at 25°C , when the concentration of $[Zn^{2+}] = 0.01\text{M}$ ($E_{Zn^{2+}/Zn}^0 = -0.76\text{V}$).

Ans: The reduction reaction can be written as $Zn_{(aq)}^{2+} + 2e \rightarrow Zn_{(s)}$

$$E = E^0 - \frac{0.0591}{2} \log \frac{1}{[Zn^{2+}]}$$

$$E = E^0 - \frac{0.0591}{2} \log \frac{1}{0.01} = -0.76 - 0.059 = -0.819\text{V}$$

Problem 1.1.2: Calculate the EMF of a Daniel cell when the concentration $[Zn^{2+}] = 0.01\text{M}$ and Concentration of $[Cu^{2+}] = 0.1\text{M}$ given that $E_{Zn/Zn^{2+}}^0 = -0.76\text{V}$ and $E_{Cu^{2+}/Cu}^0 = +0.34\text{V}$.

Ans: The cell representation is $Zn/Zn^{2+} // Cu^{2+}/Cu$

$$E_{\text{cell}}^0 = E_{\text{cathode}}^0 - E_{\text{anode}}^0$$

$$E_{\text{cell}}^0 = E_R^0 - E_L^0$$

$$E_{\text{cell}}^0 = E_{Cu^{2+}/Cu}^0 - E_{Zn^{2+}/Zn}^0$$

$$E_{\text{cell}}^0 = +0.34 - -0.76 = 1.1\text{V}$$

the cell reaction is $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$

$$E_{\text{cell}} = E_{\text{cell}}^0 - \frac{0.0591}{2} \log \frac{[Zn^{2+}]}{[Cu^{2+}]}$$

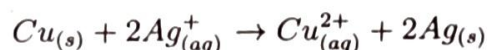
$$E_{\text{cell}} = E_{\text{cell}}^0 - \frac{0.0591}{2} \log \frac{0.01}{0.1} = 1.1 + \frac{0.0591}{2} = 1.1295\text{V}$$

Problem 1.1.3: Write the cell reaction and cell representation of a Cu-Ag cell at 25°C given that $E_{Ag^+/Ag}^0 = 0.80\text{V}$ and $E_{Cu^{2+}/Cu}^0 = 0.34\text{V}$. calculate the EMF of the cell Cu when $[Ag^+] = 0.1\text{M}$ and $[Cu^{2+}] = 0.01\text{M}$.

Ans: Since Ag^+/Ag is having higher reduction potential it undergoes reduction reaction and Cu metal undergo oxidation.



The cell reaction is obtained by equation (1) + (2) $\times 2$



The cell representation is $Cu/Cu^{2+} // Ag^+/Ag$

$$E_{\text{cell}}^0 = E_R^0 - E_L^0 = 0.80\text{V} - 0.34\text{V} = 0.46\text{V}$$

$$E_{cell} = E_{cell}^0 - \frac{0.0591}{2} \log \frac{[Cu^{2+}]}{[Ag^+]^2}$$

$$E_{cell} = 0.46 - \frac{0.0591}{2} \log \frac{0.01}{0.1^2} = 0.46 - 0 = 0.46 \text{ V}$$

Problem 1.1.4: Calculate the Equilibrium constant of Daniel cell reaction given that $E_{cell}^0 = 1.1 \text{ V}$.

Ans: Nernst equation for complete cell is

$$E_{cell} = 0.46 - \frac{0.0591}{2} \log Q$$

When the cell reaction, $Zn_{(s)} + Cu_{(aq)}^{2+} \rightarrow Zn_{(aq)}^{2+} + Cu_{(s)}$ is at equilibrium, free energy $\Delta G = 0$, ie. EMF of the cell, $E_{cell} = 0$ and reaction quotient $Q = K$ (equilibrium constant)

$$0 = E^0 - \frac{0.0591}{2} \log K$$

$$0 = 1.1 - \frac{0.0591}{2} \log K$$

$$\log K = \frac{1.1 \times 2}{0.0591}$$

$$\log K = 37.22$$

$$K = 1.67 \times 10^{37}$$

The high value of K implies that reaction taking place is almost complete.

Problem 1.1.5: A Zn rod is inserted into 0.2 M $CuSO_4$ solution at 25°C displacement reaction takes place and allowed to attain equilibrium. Find the concentration of Cu^{2+} ions at equilibrium if $E_{Zn^{2+}/Zn}^0 = -0.76 \text{ V}$ $E_{Cu^{2+}/Cu}^0 = 0.34 \text{ V}$.

Ans: The displacement reaction is $Zn_{(s)} + Cu_{(aq)}^{2+} \rightarrow Zn_{(aq)}^{2+} + Cu_{(s)}$

$$E_{cell}^0 = E_{Cu/Cu^{2+}}^0 - E_{Zn^{2+}/Zn}^0 = +0.34 - -0.76 = 1.1 \text{ V}$$

At equilibrium, free energy $\Delta G = 0$, ie: EMF of the cell, $E_{cell} = 0$, and reaction quotient $Q = K$ (equilibrium constant).

From the previous problem $K = 1.67 \times 10^{37}$. The high value of K implies that reaction taking place is almost complete. ie: concentration of Cu^{2+} at equilibrium will be very small. From equilibrium reaction

$$K = \frac{[Cu^{2+}]}{[Zn^{2+}]} = 1.67 \times 10^{37}$$

Let 'x' be the concentration of Cu^{2+} at equilibrium.

$$\frac{0.2 - x}{x} = 1.67 \times 10^{37}$$

Since 'x' is very small 'x' in the numerator can be neglected.

$$\frac{0.2}{x} = 1.67 \times 10^{37}$$

$$x = 0.119 \times 10^{-37}$$

1.1.7 Applications of Nernst Equation

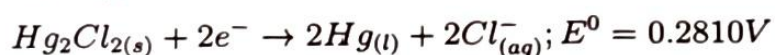
- It can be used to study the effect of electrolyte concentration on electrode potential.
- It can be used for the calculation of the potential of a cell under non-standard conditions.
- The unknown concentration of one of the ionic species in a cell can be determined with the help of Nernst Equation, provided that E_{cell}^0 and concentration of other ionic species are known.
- The pH of a solution can be calculated using this equation.
- It can also be used for finding the valency of an ion or the number of electrons involved in the electrode reaction.
- Solubility of sparingly soluble salt at any temperature can be determined.

1.2 Secondary reference electrodes

Standard hydrogen electrode is the primary reference electrode. Calomel electrode and silver/silver chloride/KCl electrodes are commonly used secondary reference electrodes. We often use secondary reference electrodes due to limitations of primary reference electrode SHE.

1.2.1 Calomel Electrode

Calomel electrode is a secondary reference electrode. This is a commonly used reference electrode since it is difficult to set up and maintain SHE. This is an example of metal/ metal insoluble salt/ common ion electrode (mercury, mercurous chloride electrode, KCl). It consists of a glass tube at the bottom of which a small amount of Hg is placed. This is covered with a paste of solid mercurous chloride (Hg_2Cl_2 or calomel) which is further in contact with a solution of KCl. A Pt wire dipped into the Hg layer is used for making electrical contact. The side tube is used for making electrical contact with a salt bridge. The electrode may be represented as $Hg/Hg_2Cl_{2(s)}/KCl$. The electrode reaction is given by



The calomel electrode can act as anode or cathode depending on the electrode potential of the coupled electrode. If the electrode potential of the coupled electrode is lower than the calomel electrode, then calomel electrode undergoes reduction and acts as cathode. If the potential of the coupled electrode is greater, calomel electrode will undergo oxidation and the reverse reaction takes place. Nernst equation of this electrode at 25°C is given by

$$E_{cal} = E_{cal}^0 - \frac{0.0591}{2} \log[Cl^-]^2$$

$$E_{cal} = 0.2810 - \frac{0.0591}{1} \log[Cl^-]$$

From the above it is seen that the potential of the calomel electrode varies with the concentration of the KCl solution which is summered in table 1.3. Saturated calomel electrode is generally used, since it is easy to setup and its emf will not change with chemical reaction.

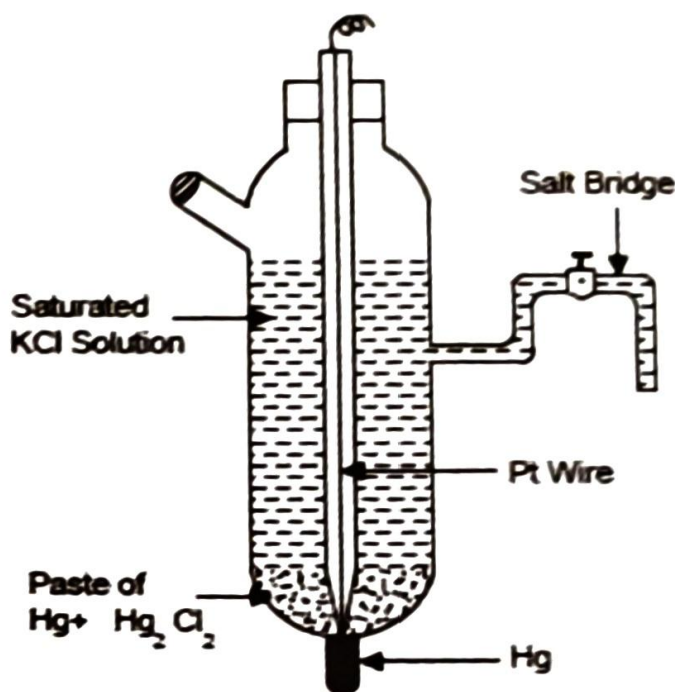


Figure 1.4: Calomel electrode

Table 1.3: Electrode potential of calomel electrode at different concentrations

KCl concentration	0.1N (DNCE)	1N (NCE)	Saturated (SCE)
Electrode potential (in Volt)	0.3335	0.2810	0.2422

Problem 1.2.1: Calculate the solubility of KCl in water at 25°C if saturated calomel electrode has a potential of 0.2422 V and standard calomel electrode has the potential of 0.2810 V.

Ans: The electrode may be represented as $\text{Hg}/\text{Hg}_2\text{Cl}_{2(s)}/\text{KCl}$. The Nernst equation of this electrode at 25°C is given by

$$E_{\text{cal}} = 0.2810 - \frac{0.0591}{1} \log[\text{Cl}^-]$$

$$0.2422 = 0.2810 - \frac{0.0591}{1} \log[\text{Cl}^-]$$

$$[\text{Cl}^-] = 4.53 \text{ M}$$

$$[\text{Cl}^-] = [\text{KCl}_{\text{saturated}}] = 4.53 \text{ M}$$

Since concentration of saturated solution is called solubility. The solubility of KCl at 25°C in water is 4.53 M.

1.2.1.1 Measurement of electrode potential using calomel electrode

It is impossible to know the absolute value of a single electrode potential because neither oxidation nor reduction takes place independently. Hence, we find the relative electrode potential by coupling it with a reference electrode. The primary reference electrode is the normal or

standard hydrogen electrode (NHE or SHE), but it is very difficult to setup. So, we often go for secondary reference electrode like saturated calomel electrode.

Zn-electrode: In order to measure the standard electrode potential of Zn electrode, (Zn- electrode dipped in 1 M $ZnSO_4$ solution), the Zn electrode is coupled with saturated calomel electrode through a salt bridge and the emf of the cell is measured.

Note: EMF is defined as the voltage of the cell when no current is drawn from the cell. Therefore, emf of the cell must be measured without drawing any current from the cell. Analog volt meter cannot be used since it draws current from the cell. Digital volt meter is used for measuring emf. In digital volt meter, the measurement of potential is based on field effect transistors, which draws only negligible current from the cell during the measurement of emf. Wheatstone bridge principle was used earlier to measure emf of cells before the advancement of digital voltmeters.

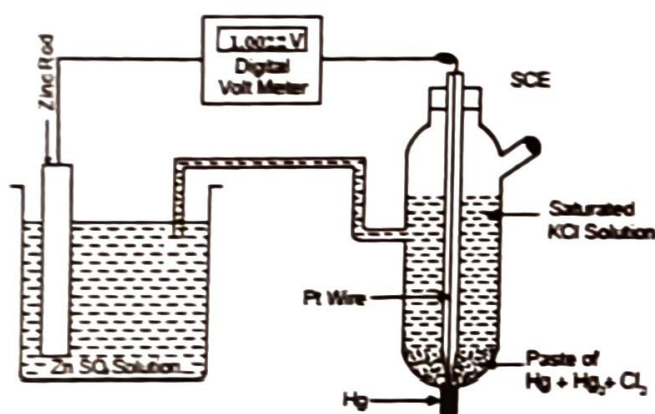
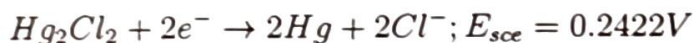


Figure 1.5: Measurement of electrode potential of Zn using saturated calomel electrode

The electrode having higher reduction potential undergoes reduction and is the cathode and the electrode having lower reduction potential undergoes oxidation and is the anode. The reduction potential of saturated calomel electrode ($E_{sce} = 0.2422\text{ V}$) is higher than standard reduction potential of Zn^{2+}/Zn (-0.76 V). Therefore, reduction reaction takes place at SCE and the Zn electrode undergo oxidation.

Cathode:



Anode:



The cell may be represented as $Zn/Zn^{2+} // Cl^- / Hg_2Cl_2 / Hg$

$$E_{cell} = E_R - E_L$$

$$E_{cell} = E_{sce} - E_{Zn^{2+}/Zn}$$

When concentration of $[Zn^{2+}] = 1\text{ M}$ and temperature = 25°C

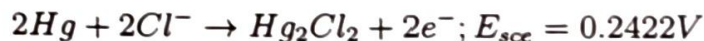
$$E_{\text{cell}} = 0.2422 - E_{Zn/Zn^{2+}}^0$$

By measuring cell emf $E_{Zn/Zn^{2+}}^0$ can be calculated.

Cu-electrode: In order to measure the standard electrode potential of Cu electrode, (Cu electrode dipped in 1 M CuSO_4 solution), the Cu electrode is coupled with saturated calomel electrode through a salt bridge and the cell emf measured.

The electrode having higher reduction potential undergoes reduction and is the cathode and the electrode having lower reduction potential undergoes oxidation and is the anode. The reduction potential of saturated calomel electrode ($E_{\text{sce}} = 0.2422\text{ V}$) is lower than standard reduction potential of $Cu^{2+}/Cu = +0.34\text{ V}$. Therefore, oxidation reaction takes place at SCE and the Cu electrode undergo reduction.

Cathode:



Anode:



The cell may be represented as $Hg/Hg_2Cl_2/Cl^-//Cu^{2+}/Cu$

$$E_{\text{cell}} = E_R - E_L$$

$$E_{\text{cell}} = E_{\text{sce}} - E_{Cu^{2+}/Cu}$$

When concentration of $[Cu^{2+}] = 1\text{ M}$ and temperature = 25°C

$$E_{\text{cell}} = E_{Cu^{2+}/Cu}^0 - 0.2422$$

By measuring cell emf $E_{Cu^{2+}/Cu}^0$ can be calculated.

1.2.2 Silver-Silver chloride electrode

The electrode can be easily constructed by placing a precipitate of AgCl on a loop of silver wire immersed in KCl solution. The electrode can be represented as $Ag/AgCl/KCl$. The electrode reaction can be written as $AgCl_{(s)} + e^- \rightarrow Ag_{(s)} + Cl^-$; $E^0 = 0.22\text{ V}$. The Nernst equation of this electrode at 25°C is given by

$$E = 0.22 - \frac{0.0591}{1} \log[Cl^-]$$

Problem 1.2.2: Find the solubility of AgCl in 1 M KCl solution if $E_{Ag/Ag^+}^0 = 0.80\text{ V}$ and $E_{Ag/AgCl/KCl}^0 = 0.22\text{ V}$.

Ans: The electrode reaction for Ag^+/Ag electrode is given by $Ag^+ + e^- \rightarrow Ag$. The Nernst equation of this electrode at 25°C is given by

$$\begin{aligned} E &= 0.80 - \frac{0.0591}{1} \log[Ag^+] \\ 0.22 &= 0.80 - \frac{0.0591}{1} \log[Ag^+] \\ [Ag^+] &= 1.54 \times 10^{-10} \end{aligned}$$

The solubility of AgCl in 1 M KCl solution is $[Ag^+] = 1.54 \times 10^{-10}$. The electrode potential of silver-silver chloride electrode is very much lower than 0.80 V due to very low concentration of Ag^+ ion in solution. Solubility product of AgCl $K_{sp} = [Ag^+][Cl^-] = 1.54 \times 10^{-10} \times 1 = 1.54 \times 10^{-10}$. The solubility AgCl in pure water can also be calculated by $\sqrt{K_{sp}} = \sqrt{(1.54 \times 10^{-10})} = 1.24 \times 10^{-5}$.

1.2.3 Glass electrode

A glass electrode is a type of ion-selective electrode made of a doped glass membrane that is sensitive to a specific ion. It consists of low melting glass having high electrical conductivity. The composition of Na_2O in this glass is slightly more than ordinary soda glass. It is the universally employed electrode for pH measurement. There are also specialized ion sensitive glass electrodes used for determination of concentration of lithium, sodium, and ammonium ions.

Construction: For pH measurement H^+ ion selective glass electrode is used. It consists of low melting glass having high electrical conductivity. The glass electrode assembly consists of a long glass tube with a thin glass bulb filled with 0.1 M HCl. The inner surface of the glass is in contact with a AgCl coated silver electrode or simply a platinum contact electrode. HCl in the bulb furnishes a constant H^+ ion concentration. For measuring the pH of a test solution, the glass bulb is immersed in the test solution such that the outer surface of the glass bulb comes in contact with the test solution. The glass membrane functions as an ion-selective resin, and an equilibrium is set up between the Na^+ ions of glass and H^+ ions in solution. The electrode reaction is $H^+ + Na_{glass}^+ \rightarrow Na^+ + H_{glass}^+$ and electrode representation is $Ag/AgCl/0.1 M H^+/glass/H^+$.

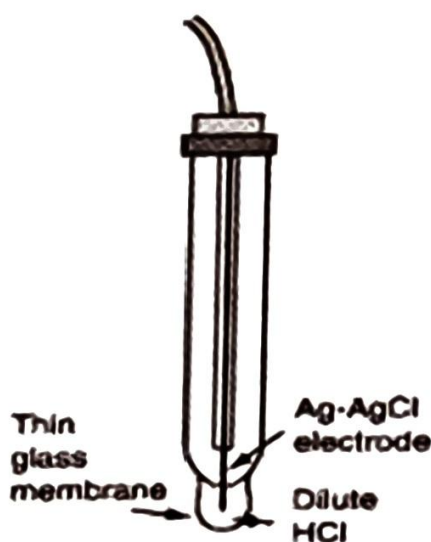


Figure 1.6: Glass electrode

The Nernst equation for the electrode can be written as

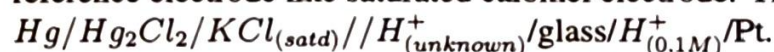
$$E_G = E_G^0 - \frac{0.0591}{1} \log \frac{1}{[H^+]}$$

$$E_G = E_G^0 - 0.0591 \times pH$$

Where E_G^0 is a constant for a particular variety of glass. $E_G^0 = 0.6$ or 0.5 depending on the variety of glass.

1.2.3.1 Measurement of pH

In order to measure the potential of glass electrode, it has to be coupled with secondary reference electrode like saturated calomel electrode. The cell can be represented as



The emf of the cell is measured and the pH can be calculated as follows

$$E_{cell} = E_{right} - E_{left}$$

$$E_{cell} = E_G^0 - 0.0591pH - 0.2422$$

$$E_{cell} = (E_G^0 - 0.2422) - 0.0591pH$$

The equation has the form $y = c + mx$, the equation for a straight line having intercept c and slope m , when we plot a graph between emf (y-axis) and pH (x-axis). The constants c and m can be determined by calibrating the pH meter with two buffer solutions of known pH. The intercept $c = E_G^0 - 0.2422$ and the slope or value of m is -0.0591 at $25^\circ C$. Its value at the experimental temperature is obtained on calibration. The pH of the test solution can be calculated by noting the E_{cell} with test solution and knowing values of c and m .

Note: Electrode assembly can be the combined type (Glass electrode and reference electrode in a single bulb) or separate (double bulb) depending on the instrument.

Problem 1.2.3: A glass electrode-calomel electrode assembly shows an emf of 208 mV with pH 4 buffer solution and - 32 mV with pH 9.2 buffer solution. Find the pH of test solution if it shows an emf of 78 mV.

Ans: The equation has the form $y = c + mx$, The equation for a straight line having intercept c and slope m when we plot a graph between emf (y-axis) and pH (x-axis).

$$208 = c + 4m \quad (1)$$

$$-32 = c + 9.2m \quad (2)$$

equation (1)-(2) gives

$$240 = -5.2m$$

$$m = -46.15$$

substituting in (1)

$$208 = c + 4 \times -46.15$$

$$c = 392.6$$

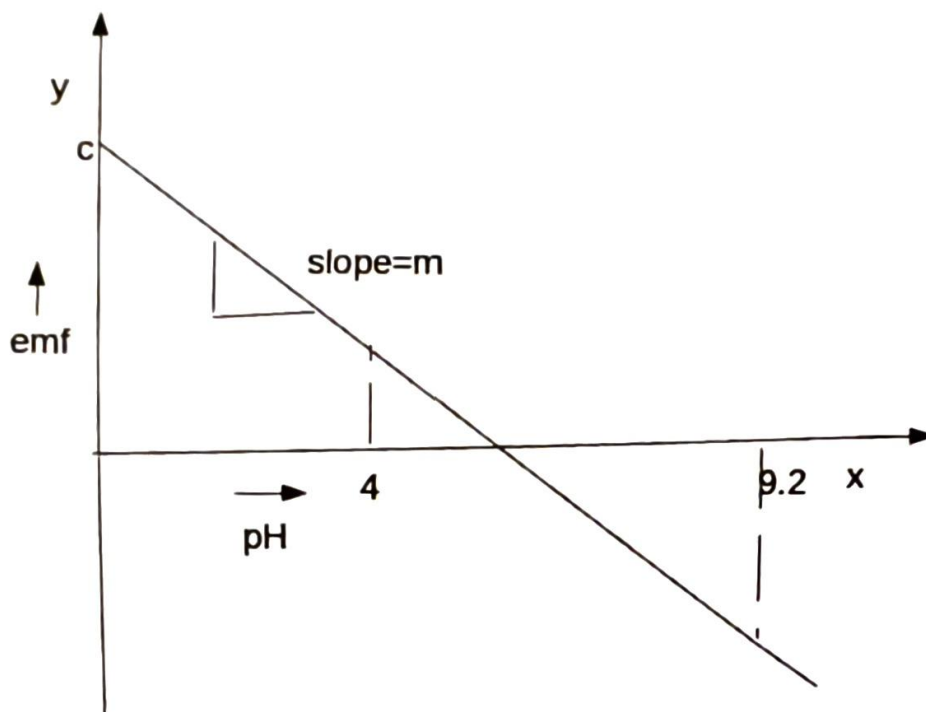


Figure 1.7: Plot of emf vs pH

emf of test solution can be written as

$$78 = c + m \times pH \quad (3)$$

Knowing the calibration constants m and c unknown pH can be determined by substituting equation (3)

$$78 = 392.6 + -46.15 \times pH$$

$$pH = 6.81$$

1.3 Conductivity

Electrical energy is taken through matter in the form of electric current. In order to have electric current, charge carriers and driving forces are essential. The charge carriers can be either electrons or ions. A battery or suitable source of electric energy can be used to provide the driving force. The substances which allow the passage of electric current through them are called conductors while those which do not allow the flow of electric current are called insulators. Conductors are of two types

1. Electronic conductors - example: metals and semiconductors.
2. Electrolytic conductors - example: salt solutions, acids, molten salts etc.

Conductivity of electrolytic conductors ($10^{-3} Scm^{-1}$) are very low when compared with metals ($10^7 Scm^{-1}$). This is due to lesser number of charge carriers per unit volume and lower mobility of ions when compared with electrons. Flow of DC current through electrolytes causes decomposition of electrolytes. While the flow of AC current has no such effects.

1.3.1 Electrical resistance (R)

Electrical resistance measures the obstruction to the flow of current. Resistance of a conductor is proportional to length (l) and inversely proportional to area of cross section (a) of the conductor.

$$R \propto \frac{l}{a}$$

$$R = \frac{\rho \times l}{a}$$

The constant of proportionality ' ρ ' is called specific resistance or resistivity of the material. Thus, specific resistance is the resistance of a conductor of unit length and unit area of cross section. The unit of resistivity is $ohm\ cm$.

1.3.2 Electrical conductance (C)

It is a measure of the ease with which the current flows through a conductor or a solution. The reciprocal of resistance is called conductance. The unit of conductance is ohm^{-1} or mho or $Siemen(S)$

$$C = \frac{1}{R}$$

1.3.3 Specific conductance or conductivity (κ)

The reciprocal of specific resistance (ρ) is called specific conductance. It is defined as the conductance of one cm^3 of a conductor or solution held between two electrodes of $1\ cm^2$ area placed on the opposite pairs of faces of cube of edge length $1\ cm$.

$$\kappa = \frac{1}{\rho}$$

$$C = \frac{1}{R} = \frac{a}{\rho \times l}$$

$$C = \frac{\kappa \times a}{l}$$

$$\kappa = \frac{C \times l}{a}$$

Where $\frac{l}{a}$ is called cell constant

$$conductivity(\kappa) = conductance(C) \times cellconstant$$

Cell constant: In a conductivity cell, the distance between two electrodes (l) and the area of electrodes are fixed. The ratio of this distance and area is a constant and is called cell constant (cell constant = $\frac{l}{a}$). Unit of cell constant is cm^{-1} or m^{-1} .

1.3.4 Conductivity cell

A conductivity cell consists of two electrode plates coated with finely divided platinum black (to minimize polarization effects). These two are welded with the ends of platinum wires that are sealed through two glass tubes. The tubes are strongly fixed such that distance between electrodes remains unaltered.

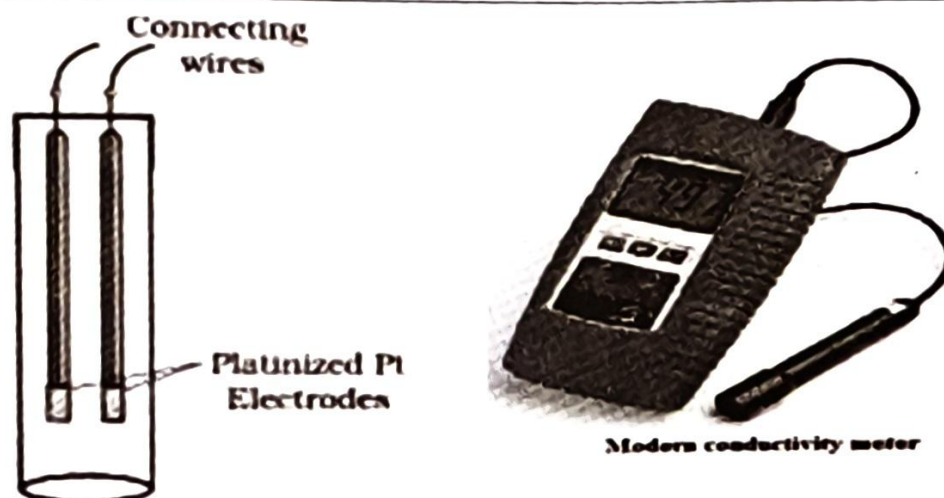


Figure 1.8: Conductivity Cell

1.3.4.1 Determination of cell constant

Cell constant may be obtained by measuring 'l' and 'a'. But it is very difficult to measure the actual area precisely in this type of small cells. In actual practice cell constant is measured by filling the conductivity cell with a solution of known conductivity like 0.1 M KCl solution. Conductivity (specific conductance) of 0.1 M KCl solution is $0.01288 \text{ S cm}^{-1}$ at 298 K. By measuring conductance of 0.1 M KCl solution in the given cell, the cell constant can be determined.

$$\text{Cell constant} = \frac{\text{conductivity of 0.1 M KCl solution}(\kappa)}{\text{conductance (C) of 0.1 M KCl solution}}$$

1.3.5 Measurement of conductivity using digital conductivity meter

Conductivity cell is connected to the digital conductivity meter and appropriate temperature is selected. Conductivity cell is then dipped into the sample solution taken in a beaker. Make sure that the cell is fully immersed in the solution. Measure the conductance of the solution by selecting the proper range switch (milli S or micro S). Digital conductivity meter directly reads out the conductance. In digital conductivity meter 5000 Hz Ac is used to measure conductance.

Conductivity of the solution can be calculated using the equation,

$$\text{Conductivity} = \text{conductance} \times \text{cell constant.}$$

Problem 1.3.1: Calculate the conductivity of given sample of water at 298 K, which shows a conductance of $560 \mu\text{S}$ in the given cell at 298 K. A standard solution of 0.1 M KCl shows a conductance of 12.34 mS in that cell. (Given that conductivity of 0.1M KCl at 298 K is $0.01288 \text{ S cm}^{-1}$.)

Ans

$$\text{Cell constant} = \frac{\text{Conductivity of KCl}(\kappa)}{\text{Conductance of KCl}(C)}$$

$$\text{Cell constant} = \frac{0.01288 \text{ Scm}^{-1}}{12.34 \times 10^{-3} \text{ S}} = 1.043 \text{ cm}^{-1}$$

$$\begin{aligned} \text{Conductivity of given sample of water}(\kappa) &= \text{conductance}(C) \times \text{cellconstant} \\ &= 560 \mu\text{S} \times 1.403 \text{ cm}^{-1} = 583 \mu\text{Scm}^{-1} \end{aligned}$$

1.4 Types of electrochemical cells

Commercial cells which are used as sources of electrical energy are of three main types: Primary cells, secondary cells and fuel cells.

1.4.1 Primary cells

Primary cells are based on cell reactions which are not reversible. Once the cell reaction is complete, the cell is discharged and cannot be recharged again e.g. Daniel cell, Weston Cd cell, Leclanche cell (dry cell) etc.

1.4.2 Secondary cells (Storage cells or Accumulators)

They are galvanic cells in which the cell reaction can be reversed by applying an external source of current. These can be discharged and recharged many times until the electrode materials last, eg: Lead - acid battery, Ni - Cd battery, Li - ion cell etc. Research are progressing for the development of batteries having high energy density, high cyclic life, explosion free and low cost for electric vehicles and solar grid storage. Apart from the development of various types of lithium ion batteries, sodium ion, aluminium ion and organic radical batteries are also progressing.

The term battery is normally used to denote a number of galvanic cells connected in series. When cells of the same type are connected in series, their emf get added. When they are connected in parallel, emf remain the same as that of a single cell but internal resistance of the combination gets halved. Thus, more current can be drawn from the cell without much drop in potential. Internal resistance of a cell depends on electrode plate area and cell chemistry. When electrode plate area increases internal resistance decreases. When size of the battery increases the energy storage increases. Energy storage of the battery is expressed in ampere-hour (Ah)

1.4.2.1 Lithium-ion cell (Li-ion battery)

Li-ion cell is a rechargeable battery in which Li-ion moves between anode and cathode during charging and discharging. It is more popular than other batteries because of the various characteristics of lithium. Li has a higher negative value for reduction potential and a lower atomic mass than any other metal. Only 6.94 g of Li is needed to provide 1 mole of electrons. Lithium-ion batteries are considered to be the most suitable material for the hybrid electric and fully electric vehicles.

Construction of Li-ion cell: In typical lithium-ion cell, the positive electrode is made by using a crystalline mixed oxide LiCoO_2 or $\text{Li}_2\text{O} \cdot \text{Co}_2\text{O}_3$. The crystal contains both Li^+ ions and Co^{3+} ions occupying the voids of the oxide array. Such crystals are usually synthesized by hydrolysis of the respective metal salts under hydrothermal condition. The negative terminal

is made of graphite. The electrolyte is a solution of Li-salt ($LiPF_6$, $LiBF_4$ or $LiClO_4$), in an organic solvent such as ether or a solid phase polymer electrolyte which can transport Li^+ ion. A solid electrolyte separator (SEI) separates the anode and cathode compartments.

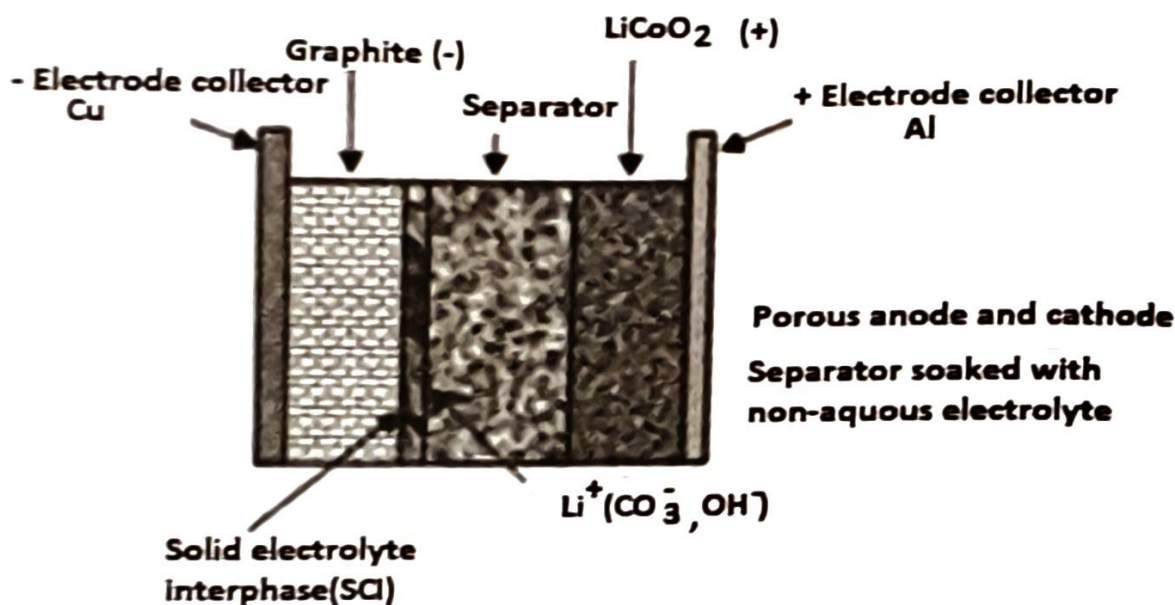


Figure 1.9: Construction of Li-ion cell

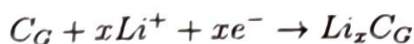
Charging: During charging the cell acts as an electrolytic cell, the electrical energy is converted into chemical energy.

Anode (+):



Here $x < 1$ mole, say 0.9 (or 90 percentage) i.e., some Li^+ ions get removed from the crystal lattice, and in the respective lattice sites more stable Co^{3+} ion get oxidized to less stable Co^{4+} and electrical neutrality is maintained in the lattice.

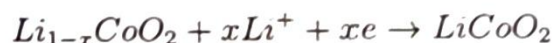
Cathode (-): The Li^+ ions produced at the anode get migrated through the electrolyte and get reduced to lithium atom at the graphite electrode



Li atoms are inserted between the layers of carbon atoms. This so called lithiated graphite is written as Li_xC_G , containing no Li metal as such. This is why Li-ion battery is not called as Li battery. When Lithium-ion cells are allowed to charge completely ($x=1$) anodic material becomes $Li_{1-1}CoO_2$ or CoO_2 , which is an explosive material (due to liberation of O_2 gas on decomposition). Therefore, automatic built-in circuit breakers are there inside battery which allows charging only up to 90 percentage ($x = 0.9$).

Discharging: During discharging, the cell act as an electrochemical cell, the chemical energy is converted to electrical energy. When the cell act as an electrochemical cell, $LiCoO_2$ acts as cathode while the graphite acts as anode.

Cathode(+):



Anode(-):



Table 1.4: Different types of anodic materials

Anodic Material	Average Voltage	Gravimetric Capacity
$LiCoO_2$	3.7 V	140 mAh/g
$LiMnO_2$	4.0 V	100 mAh/g
$LiNiO_2$	3.5 V	180 mAh/g
$LiFePO_4$	3.3 V	150 mAh/g

Lithium-ion cell based on $LiCoO_2$ has an EMF of 3.7 V. The cell is having high energy to weight ratio and has high cyclic life. They are widely used in cell phones, laptop computers, digital cameras etc., due to their long cycle life and high energy density. The cell is not suitable for making large size battery because of possibility of explosion. Another type of Li-ion cell has been developed based on the active materials like, $LiFePO_4$. The cell is having a potential of 3.3 V. The cell is suitable for making large size battery suitable for automobile application. Research indicate that lithium iron phosphate batteries are more stable and safer. More stable lithium nickel oxides were also developed in recent years.

The most popular cathode material is graphite; every 6 carbon atoms in graphite can accommodate a lithium atom. The materials like lithium doped titanium dioxide, hard carbon, tin / cobalt alloy and silicon/carbon nano-materials were also been developed as cathode material. When Lithium doped titanium dioxide is used instead of graphite, during charging Li^+ ions are transported through the electrolyte and get into the lattice, in the corresponding lattice more stable Ti^{4+} get reduced to less stable Ti^{3+} to maintain the electrical neutrality of the lattice.

Advantages of Li-ion batteries

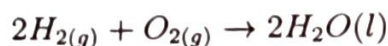
- Good cycle stability
- High energy density
- Low self-discharge rate
- No memory effect

1.4.3 Fuel Cells

A fuel cell is a galvanic cell in which the chemical energy of a fuel is directly converted to electrical energy. It is an electrochemical conversion device i.e.; it produces electricity by converting the chemical energy of a fuel and an oxidant, which reacts in the presence of an electrolyte and not by combustion. It differs from ordinary electrochemical cell in the sense that, here the reactants are supplied from outside rather than forming an integral part of the cell and the energy should be withdrawn indefinitely from a fuel cell as long as the outside supply of the fuel is maintained. It is similar to the electricity generator set, which converts the chemical energy of petrol and air into electricity. It is a power generator rather than a rechargeable battery. A fuel cell can be generally represented as Fuel/-ve electrode/electrolyte/+ve electrode/oxidant. The most successful and important fuel cell is hydrogen-oxygen fuel cell.

1.4.3.1 Hydrogen-Oxygen Fuel Cell (acid electrolyte)

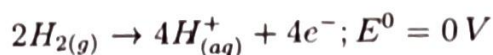
Hydrogen-oxygen fuel cell produces electricity based on the following exothermic reaction



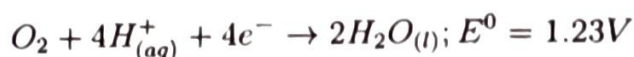
Here hydrogen acts as the fuel and oxygen acts as the oxidant, hence its simply known as hydrogen fuel cell.

It consists of two electrodes made up of porous graphite, impregnated with metal catalysts such as platinum. Arrhenius acid like phosphoric acid is generally employed as the electrolyte. A fabric like matrix made of teflon bonded silicon carbide dispersed with highly concentrated phosphoric acid (H_3PO_4) acts as the electrolyte. This electrolyte matrix transports the hydrogen ions (since phosphoric acid is used as the electrolyte its also known as phosphoric acid fuel cell). Hydrogen and oxygen gas under pressure (50 atm) are continuously passed through the anodic and cathodic chambers respectively. The cell operates at about a temperature of 150 - 200°C, since the ionic conductivity of phosphoric acid is very low at low temperatures. Heat is generated during the reaction of hydrogen with oxygen, and so cooling plates are used to maintain a cell operating temperature of about 200°C. Schematic representation of the cell is shown in fig. 1.10. The electrode reactions are as follows

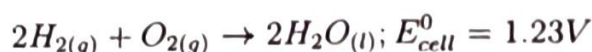
At anode: Hydrogen is oxidized to H^+ ions at the anode and the electrons formed will flow through the external circuit to the cathode.



At cathode: Oxygen is getting reduced to water at the cathode by consuming the H^+ ions from the electrolyte. H^+ ions from the electrolyte are transported to the oxygen compartment and combines with oxygen gas and electrons to form water. H^+ ion deficiency at the electrolyte side is balanced by the H^+ ion formed at the anode. Water formed is drained out through the oxygen compartment side.



Net reaction is



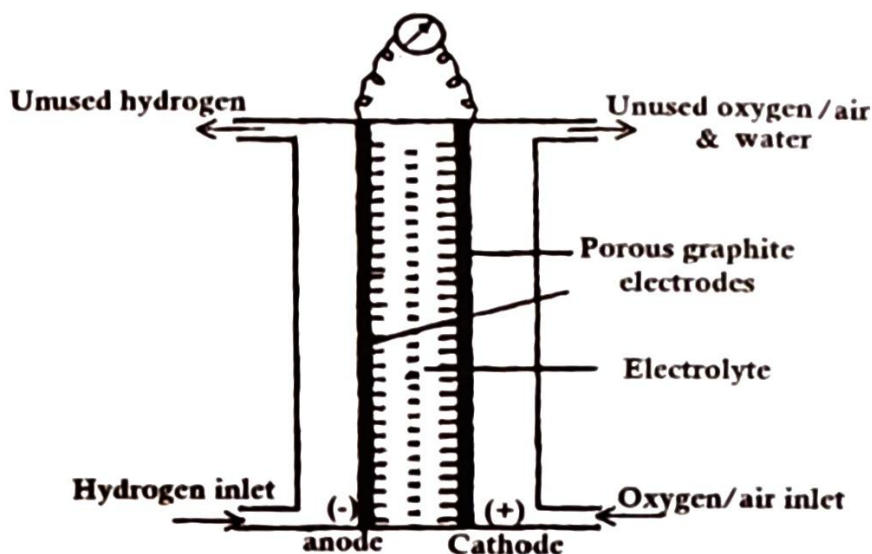


Figure 1.10: Hydrogen-oxygen fuel cell

These type of fuel cells are mainly used for stationary power generation and military applications due to their stable operation and long lifetime.

There is another type of hydrogen fuel cell which works at relatively low temperature (60 - 80°C), which uses perfluorinated sulfonic acid membrane (proton exchange membrane) as the electrolyte. Nafion, introduced by du Pont is one of the most advanced commercially available perfluorinated sulfonic acid membrane. Here also the reactions at the electrodes are same as that discussed above. Since it operates at relatively low temperature less warm-up time is needed and starts quickly. Hence, these cells are suitable for use in vehicle applications including cars, buses, and heavy-duty trucks.

Hydrogen-oxygen fuel cell has also been developed in alkaline medium which also gives the same emf. Unlike alkaline fuels cell, acid fuel cells are tolerant to CO and CO_2 which allows CO and CO_2 in the hydrogen and air feed, as they do not react in the acidic medium.

There are other types of fuel cells available, such as Solid-Oxide Fuel Cells (which uses solid ceramic electrolyte) and molten carbonate fuel cells (which uses molten metal carbonate electrolyte) which operate at relatively higher temperatures (500 - 1000°C). Since they operate at higher temperatures water formed can be easily removed as steam. Other fuels such as methanol, methane, hydrazine etc. and oxidants such as air, chlorine, chlorine dioxide etc. are also being used.

Electricity generated from a single cell is only enough for the smallest applications. Therefore, fuel cell stack which contains hundreds of individual fuel cells combined in series is generally used.

Advantages of hydrogen - oxygen fuel cell

1. The fuel cell has high efficiency and can convert about 80% of available chemical energy to electrical energy.

2. It does not cause any pollution since the only byproduct is water and they are free of noise, vibration, heat transfer etc.
3. Used in space craft due to its light weight and the byproduct water is used for drinking in space stations.
4. It can supply energy continuously as long as the reactants are fed.
5. The concentration of the electrolyte remains unchanged.
6. It is simple and easy to operate.

Note: Commercial vehicles which utilize hydrogen as the fuel are now in use (eg: Toyota Mirai, Hyundai Nexa). Various cities around the world have begun to incorporate hydrogen fuel cell buses in their public transport. First hydrogen fuel cell bus in India was launched in September 2023. This demands the production of more hydrogen, preferably in an ecofriendly way. Nowadays, solar energy production is becoming increasingly efficient and widespread. The surplus electricity produced (after consumption) by the solar cell during the day time can be used to electrolyze water into hydrogen and oxygen, which can be stored. This hydrogen can be used by the fuel cell to produce electricity, which helps to improve hydrogen economy through sustainable approach.

Disadvantages

1. There is difficulty in getting a suitable electro-catalyst which can function efficiently for long periods without getting contaminated and deteriorated.
2. Gaseous fuels are difficult to store and handle.
3. The electrolyte used is highly corrosive.
4. Fuels used must be extremely pure.
5. The technology is expensive.

1.5 Corrosion

Corrosion is the process of disintegration or deterioration of a metal by chemical or electrochemical reaction with the environment eg: rusting of iron and tarnishing of silver, copper etc.

Cause of corrosion: Most metals exist in nature as oxides, sulfides, carbonates, chlorides, silicates etc. because these compounds represent their thermodynamically stable state. The pure metals are extracted from these ores after expending a lot of energy. The pure metal will always then be at a higher energy state than its ore. Thus, the metal which is now in an excited state will have a natural tendency to revert back to its thermodynamically stable state. This is the basic reason for metallic corrosion. However, metals like gold, platinum etc. which exist in the elemental state in nature have excellent corrosion resistance in natural environment.

Theories of Corrosion: Based on the mechanisms, corrosion can be explained by two main theories.

1. Dry or chemical corrosion

- (a) Oxidation corrosion (High temperature corrosion of iron)
- (b) Corrosion by other gases (oxidation by gases like Cl_2)
- (c) Liquid metal corrosion (Dissolution of metals in liquid metals like mercury, liquid sodium etc.)

2. Wet or electrochemical corrosion

- (a) Hydrogen evolution type corrosion
- (b) Oxygen absorption type corrosion

1.5.1 Mechanism of Wet or Electrochemical corrosion

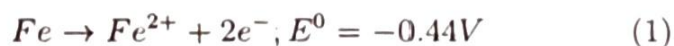
It takes place under moist conditions through the formation of galvanic cells. When a conducting liquid is in contact with a metal or when two dissimilar metals are either immersed or dipped partially in a solution. This type of corrosion involves the following steps

1. Oxidation occurs at the anode generating metallic ions; thus, corrosion occurs at the anode.
2. Reduction takes place at the cathode. Here the dissolved constituents like O_2 or H^+ in the conducting medium accept the electron and form non-metallic ions like (OH^-) .
3. Diffusion of metallic and non-metallic ions towards each other through the conducting medium and formation of corrosion product somewhere between anodic and cathodic areas.
4. Separate anodic and cathodic areas are formed due to the deposition of corrosion products between which current flows through the conducting medium.

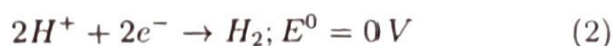
1.5.1.1 Hydrogen evolution corrosion in acid medium

This occurs when metals are in contact with acidic solution in the absence of sufficient oxygen concentration, eg., Fe or Al in contact with HCl. Hydrogen evolution corrosion takes place in the case of iron immersed in acid medium; this is due to the fact that hydrogen electrode potential in acid medium ($E_{H^+/H_2}^0 = 0$) is higher than electrode potential of iron ($E_{Fe^{2+}/Fe}^0 = -0.44V$)

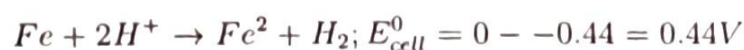
Anode



Cathode



Net reaction (1)+(2)



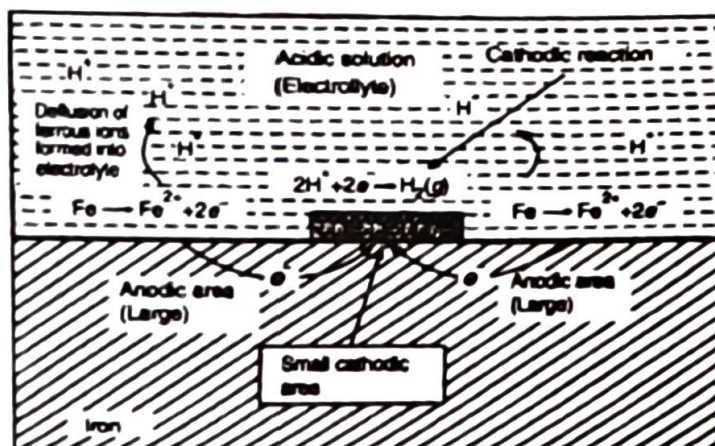


Figure 1.11: Hydrogen evolution corrosion

In this type of corrosion, displacement of H^+ from acidic solution by the metal takes place. The metals having higher negative reduction potentials like Fe, Zn, Al etc. will be anodic and suffer corrosion, while the metals having positive reduction potentials like Cu, Ag etc. will be cathodic and get some galvanic protection.

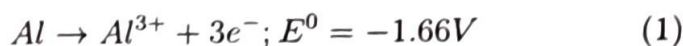
1.5.1.2 Hydrogen evolution corrosion in alkaline medium

Hydrogen evolution corrosion does not take place in the case of iron immersed in alkaline medium. This is due to the fact that hydrogen electrode potential in alkaline medium $E_{H_2/OH^-}^0 = -0.828 \text{ V}$ is lower than electrode potential of iron ($E_{Fe^{2+}/Fe}^0 = -0.44 \text{ V}$).

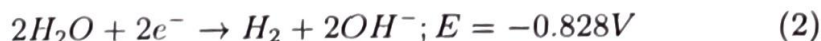
Note: The variation of potential with pH for hydrogen electrode is given by $E = -0.0591\text{pH}$, for alkaline solution pH can be taken as 14 and therefore $E = -0.0591 \times 14 = -0.828 \text{ V}$.

Metals like aluminum can give hydrogen evolution corrosion in alkaline medium also. This is due the fact that electrode potential hydrogen electrode in alkaline medium (-0.828 V) is higher than aluminum ($E_{Al^{3+}/Al}^0 = -1.66 \text{ V}$).

Anode



Cathode

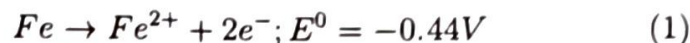


Here corrosion cell potential is

$$E_{cell} = E_{cathode} - E_{anode} = -0.828 - (-1.66) = 0.832V$$

1.5.1.3 Oxygen absorption corrosion in acid medium

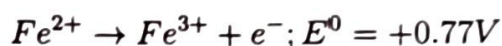
This occurs when metals are in contact with water having dissolved oxygen in acidic or even neutral environment eg; rusting of iron.

Anode**Cathode**

Here corrosion cell potential in acid medium is

$$E_{cell}^0 = E_{cathode} - E_{anode} = 1.23 - -0.44 = 1.67V$$

Fe^{2+} produced here can further oxidized to Fe^{3+} in acidic or neutral environments, since oxygen electrode potential is greater than $E_{Fe^{3+}/Fe^{2+}}^0 = +0.77 V$.



Fe^{3+} produced here forms an insoluble precipitate of $Fe(OH)_3$ called yellow rust. If the amount of moisture is less black rust $FeO.Fe_2O_3$ (magnetite Fe_3O_4) is formed. The rust formation leads to intense localized corrosion called pitting corrosion. This is due the formation of separate cathodic and anodic area. The precipitate adhered area not in contact with oxygen is called anodic area. In this area iron oxidizes and pits are formed. Oxygen exposed area is called cathodic area (fig 1.12).

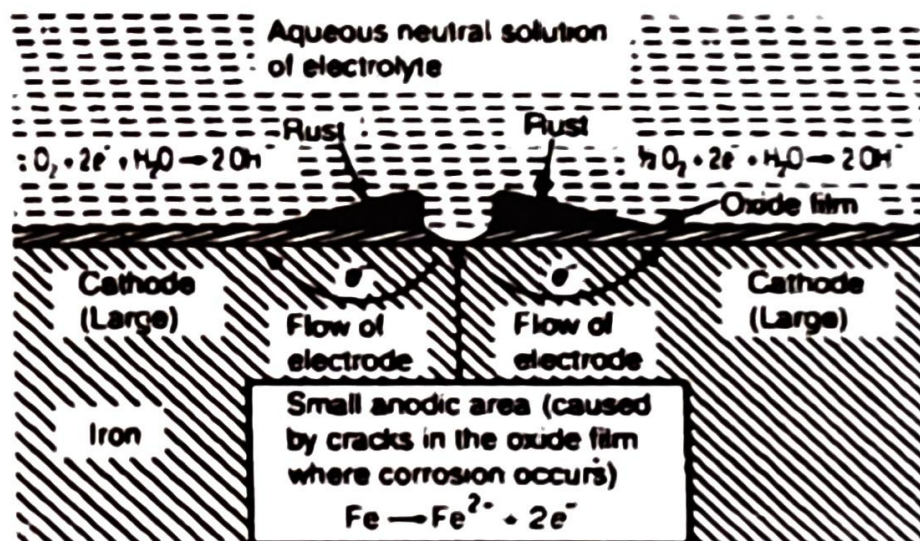
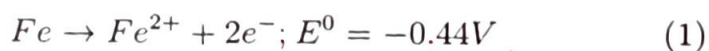


Figure 1.12: Oxygen absorption corrosion

1.5.1.4 Oxygen absorption corrosion in alkaline medium

This occurs when metals are in contact with water having dissolved oxygen in alkaline environment.

Anode**Cathode**

Here corrosion cell potential in acid medium is

$$E_{cell} = E_{cathode} - E_{anode} = 0.40 - -0.44 = 0.84V$$

Note: The variation of potential with pH of oxygen electrode is given by $E = 1.23 - 0.0591pH$, for alkaline solution pH can be taken as 14 and therefore $E = 1.23 - (0.0591 \times 14) = +0.40V$.

Fe^{2+} produced here cannot further oxidized to Fe^{3+} in alkaline environments, since oxygen electrode potential is less than ($E_{Fe^{3+}/Fe^{2+}}^0 = +0.77V$) in alkaline medium. Here no insoluble precipitate of $Fe(OH)_3$ is produced, thus no intense localized corrosion.

Note: The reinforcing steel bars used in concrete slabs do not corrode heavily by pitting due to the alkalinity of cement, but after 80 to 100 years the alkalinity of cement is lost by the absorption of carbon dioxide and leads to failure of concrete structures.

1.5.2 Galvanic Series

The electrochemical series gives useful information regarding the reactivity of metals, but it does not provide sufficient information in predicting the corrosion behavior. In electrochemical series, the metals having lower reduction potential in the series are more anodic and undergo corrosion faster. However, there are exceptions like Ti and Al. As per electrochemical series it is expected that Ti corrodes faster than Ag, but Ti is found to be less reactive than Ag. Also, Al is expected to corrode faster than Zn, but found to be less reactive than Zn. This is because Ti and Al develops a highly protective oxide film on their surface and hence their effective E^0 is more positive (or less negative). Therefore a more practical series have been prepared by studying the corrosion of metals and alloys in unpolluted sea water by measuring their oxidation potential using calomel electrode reference electrode. Galvanic series is a series in which the metals and alloys are arranged in the order of their corrosion tendencies. A metal high in the series is more anodic and undergo corrosion faster (fig.1.13). The differences between galvanic and electrochemical series are summarized in table 1.5.

Corrosion current: Corrosion current per unit area is another parameter to be measured for judging corrosion rate of metals and alloys. It is measured by coupling the metal with inert electrode like graphite and immersed in seawater (or 4 percentage NaCl solution).

Mg, Mg alloys, Zn, Al, Cd, Mild steel, Cast iron, solder, Pb, Sn

Anodic (Easily corroded)

**Brass,
Cu,
Ni**



Cathodic (protected) Pt, Au, Ti, Ag, Cr, stainless steel

Figure 1.13: Galvanic series

Table 1.5: Comparison between electrochemical series and galvanic series

Electrochemical series	Galvanic series
E^0 is measured for metals and non-metals.	E^0 is measured for metals and alloys.
E^0 is measured by dipping pure metals in their 1M salt solution.	E^0 is measured by dipping metals and alloys in unpolluted sea water.
SHE is used as reference electrode.	Calomel electrode is used as reference electrode.
Position of a metal is fixed.	Position of metal and alloy may change.
It predicts the relative displacement tendencies.	It predicts the relative corrosion tendencies.

1.5.3 Cathodic protection

The principle involved in this method, is to force the metal to be protected to behave as the cathode so that corrosion is prevented. There are two types of cathodic protection.

1. **Sacrificial anodic protection or galvanic protection:** Here the metallic structure (eg., Underground pipe lines, cables, ship hulls etc.) to be protected is connected by a wire to a more anodic metal. So, the anodic metal sacrifices and undergoes corrosion. Metals like Mg, Zn and Al and their alloys are used as sacrificial anodes.

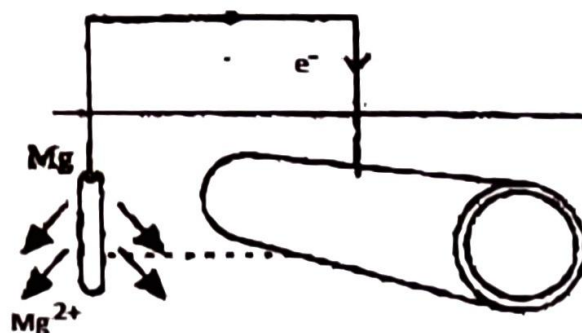


Figure 1.14: Sacrificial anodic protection

2. **Impressed current cathodic protection:** In this method an impressed current is applied to convert the corroding metal from anode to a cathode. A direct current from a DC source is applied to an anode (inert material like graphite) which is buried inside the

soil and to the corroding metal structure to be protected. A DC voltage applied such that corrosion current become zero or -ve. Ships docked in harbour are protected by this method.

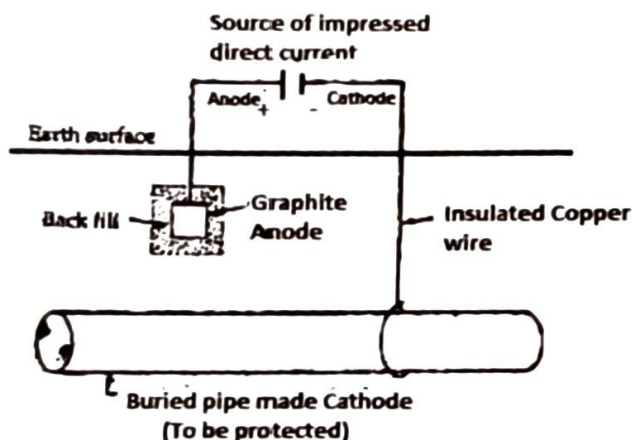


Figure 1.15: Impressed current cathodic protection

1.5.4 Electroplating

Electroplating is the most important and the most common industrial method for producing metallic coatings. This is the process by which the coating metal is deposited on the base metal by passing a direct current through an electrolytic solution, containing the soluble salt of the coating metal. The film so produced is quite uniform.

Theory of electroplating: The object to be plated (base metal) is made as the cathode of the electrolytic cell whereas the anode is either the coating metal itself or an inert material of good electrical conductivity like graphite. The electrolyte is a soluble salt solution of the coating metal. When DC current is passed through the electrolyte electrolysis occurs and the metal is coated on the surface of the substrate.

Objectives of electroplating

- To increase the resistance to corrosion
- To increase the resistance to chemical attack and wear resistance
- To improve the physical appearance and hardness
- To improve the surface properties
- To increase the decorative and commercial values of metal

1.5.4.1 Copper electroplating:

The article to be electroplated is first treated with organic solvent to remove oil, grease etc. Then it is made free from surface scales, oxides etc. by treating with dil.HCl or dil. H_2SO_4 .

The cleansed article is then made as the cathode of the electrolytic cell. Pure copper metal rod is taken as the anode. Both the electrodes are dipped in copper sulphate solution (electrolyte) taken in an electroplating tank (fig. 1.16). Direct current is passed through the electrolyte, electrolysis occurs and Cu^{2+} ions move towards the cathode and gets deposited there.

Cathode(–)



The free SO_4^{2-} ions migrate to the copper anode and dissolve an equivalent amount of copper.

Anode(+)

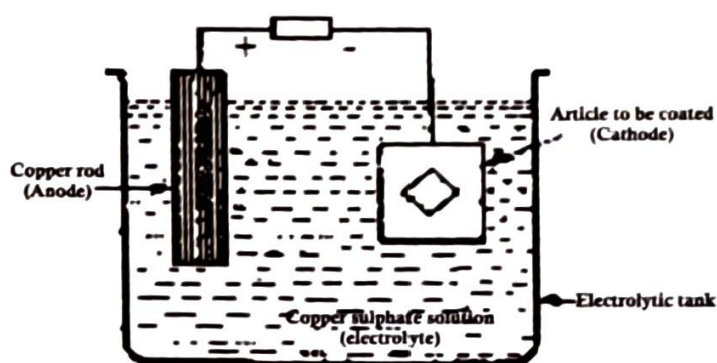
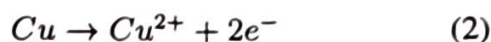
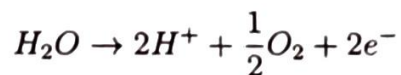


Figure 1.16: Copper electroplating

If the anode is made of some inert material like graphite, then electrolytic salt is added continuously in order to maintain proper coating metal ion concentration in the electrolytic bath. Moreover, the process of electroplating goes on non-stop, since the anode is not consumed and its replacement is not required. When inert electrode is used, anode reaction can be represented as



For brighter and smooth deposits, favorable conditions such as low temperature, medium current density and low metal ion concentration are used.

1.5.4.2 Applications of copper electroplating

- Steel articles used in automobile industry are generally given three successive coatings of Cu, Ni and Cr. This is done for corrosion protection and to impart shining and clear metal appearance.
- Copper electroplating is used in decorative items, kitchen wares, jewelry etc. to enhance the aesthetic appearance.

1.5.5 Electroless plating

Electroless plating is defined as the controlled autocatalytic deposition of a continuous layer of a noble metal from its salt solution on a catalytic surface of the base metal by using a suitable reducing agent without using electrical energy. This method allows the easy preparation of thin films using simple equipment.

Principle: Electroless deposition is an autocatalytic process where the substrate develops a potential in a plating bath containing metallic ions, reducing agent, complexing agent, stabilizer and other components. This is a non-galvanic plating method that involves several simultaneous reactions in an aqueous solution, which occur without the use of external electrical power. The reaction is accomplished when hydrogen is released by a reducing agent, normally sodium hypophosphite (the hydrogen leaves as a hydride ion), and oxidized, thus producing a negative charge on the surface of the metal or part. The key requirement of an electroless deposition bath of this type is to arrange the chemistry such that the kinetics of homogeneous electron transfer from the reducing agent to the metal ion is slow. This is essential because otherwise the metal ion would simply be reduced in the bulk solution. A catalyst that accelerates the rate of metal ion reduction is thus applied to the surface to be coated. In this way, metal ions are reduced only at the surface and the surface becomes coated with the desired metal.

The stability of electroless plating fully depends on the substrate material, the pretreatment process, the type of solution used, and the pH and temperature during plating.

Process of electroless plating: The process of electroless plating involves the following steps.

1. Preparation of the surface to be plated: In this step the surface is activated by following ways
 - (a) Etching- Removal of impurities by acid treatment.
 - (b) Electroplating - A thin layer of the metal to be plated or any other suitable metal is coated on the surface of the object.
 - (c) Chemical Treatment - The plastic objects and printed circuit boards are treated with SnCl_2 and PdCl_2 . This treatment yields a thin layer of Pd on the treated surface.
2. Preparation of plating bath: The plating bath should include the following
 - (a) Soluble salt of the metal (like chloride or sulphate)
 - (b) Complexing agents like EDTA, tartrate, citrate etc. to prevent excess deposition.
 - (c) Reducing agents like sodium thiosulphite, formaldehyde etc.
 - (d) Exaltants or accelerators like succinates, glycinate, fluorides etc. to enhance the plating rate.
 - (e) Stabilizers like calcium ions, lead ions, thiourea etc. to prevent decomposition of the bath and to impart stability to the solution.
 - (f) Buffer solution like Rochelle salt, Boric acid to maintain pH.

The article to be plated is immersed in the bath containing the metal salt and reducing agent. The plating is carried out in a series of tanks. The rate of deposition is controlled by the amount of reducing agent and the type of chelating agent.

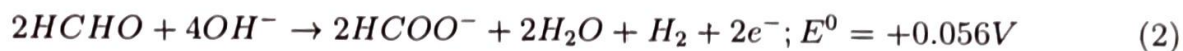
Advantages:

1. Surface to be coated does not need to be conductive, thus electroless plating can be made on both metals and plastics.
2. Electrical power and other accessories are not required.
3. Gives uniform coating even on the objects with irregular shapes, holes recesses etc.
4. Coating provides better wear resistance.
5. The coating is less porous and provides better corrosion resistance.
6. Provides flexibility in plating volume and thickness.

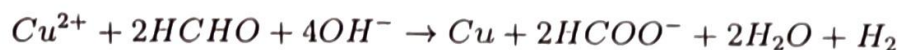
1.5.5.1 Electroless Copper plating

Electroless copper plating is used to deposit a layer of copper on a substrate without the use of electrical energy. It is commonly used for making printed circuit boards.

The object to be plated is degreased using organic solvent followed by acid treatment. This object is then immersed in plating bath containing cupric salt, reducing agents, buffer and complexing agent.

Cathode**Anode**

Net reaction (1)+(2)



Electroless copper solutions, utilizing formaldehyde as reducing agent, employ high pH, above 12 (typical NaOH concentration is > 0.1N; (pH above 13)).

1.5.5.2 Applications of electroless copper plating

1. This type of coating is mainly used in double or multilayered printed circuit boards in which plating through holes is required.
2. This is used as base coating for subsequent electroplating.

1.6 Solved problems

Problem 1.6.1: Calculate the single electrode potential of Zn^{2+}/Zn electrode, when the concentration of $Zn^{2+} = 0.1$ M and temperature at $50^\circ C$.

Ans: Reduction reaction is $Zn^{2+} + 2e^- \rightarrow Zn$

$$E = E^0 - \frac{2.303RT}{nF} \log \frac{1}{[Zn^{2+}]}$$

$$E = 0.76 - \frac{2.303 \times 8.314 \times 323}{2 \times 96500} \log \frac{1}{0.1}$$

$$E = -0.792V$$

Problem 1.6.2: Find out the pH of a solution measured using a hydrogen electrode, which is coupled with a saturated calomel electrode. The emf of the combined cell is 0.523 V at $25^\circ C$. (given that $E_{sce} = +0.2422$ V).

Ans: Since SHE ($E_0 = 0$) has lower potential than SCE, Hydrogen electrode undergo oxidation, the cell can be represented as $Pt/H_2/H^+_{(c=?)} // Cl^-_{(saturated)} / Hg_2Cl_2/Hg$

$$E_{cell} = E_R - E_L$$

$$E_{cell} = E_{SCE} - E_{H^+/H_2}$$

but for hydrogen electrode, the half-cell reaction is $H^+ + e^- \rightarrow \frac{1}{2}H_2$. The variation of potential with pH is given by

$$E_{H^+/H_2} = 0 - 0.0591 \log [H^+] = -0.0591 pH$$

$$\text{Therefore } E_{cell} = E_{SCE} - (-0.0591 pH)$$

$$0.523 = 0.2422 + 0.0591 pH$$

$$pH = 4.74$$

Problem 1.6.3: Find out the pH of a solution in which a glass electrode is dipped and is coupled with a saturated calomel electrode. The emf of the combined cell is 0.205 V at $25^\circ C$ (given $E_{sce} = +0.2422$ V and $E_{glass}^0 = 0.57$ V).

Ans: The cell can be represented as $Hg/Hg_2Cl_2/KCl_{(satd)} // H^+/glass/H^+/Pt$

$$E_{cell} = E_{Right} - E_{Left}$$

$$E_{cell} = (E_G^0 - 0.0591 pH) - 0.2422$$

$$0.205 = 0.57 - 0.0591 pH - 0.2422$$

$$pH = 2.07$$

Problem 1.6.4: (a) Find the potential of hydrogen electrode at 25°C for solution of $\text{pH} = 0$ and $\text{pH} = 14$

(b) Aluminium metal when reacts with acid and base liberates H_2 gas; Whereas Fe metal can liberate H_2 only from acids. Give reason based on potentials given, $E_{\text{Al}^{3+}/\text{Al}}^0 = -1.66\text{ V}$ $E_{\text{Fe}^{2+}/\text{Fe}}^0 = -0.44\text{ V}$.

Ans:

(a) For hydrogen electrode at 25°C , $\text{H}^+ + \text{e}^- \rightarrow \frac{1}{2}\text{H}_2$

$$E_{\text{H}^+/\text{H}_2} = 0 - 0.0591 \log[\text{H}^+] = -0.0591 \text{pH}$$

$$\text{when pH}=0 \text{ (1 M HCl)} \quad E = -0.0591 \times 0 = 0$$

$$\text{when pH}=14 \text{ (1 M NaOH)} \quad E = -0.0591 \times 14 = -0.829\text{ V}$$

(b) Both Aluminium and Iron can liberate H_2 from acids since hydrogen electrode potential in 1 M acid ($\text{pH} = 0$) is zero which is greater than Al^{3+}/Al potential (-1.66V) and Fe^{2+}/Fe potential (-0.44V). Only aluminium can liberate H_2 from basic medium since H_2 electrode potential at $\text{pH} = 14$ is -0.82 V which is greater than Al^{3+}/Al potential of -1.66V .

Problem 1.6.5 Which medium in acidic oxygen rich or alkaline oxygen rich medium corrosion of iron is rapid? Given that $E_{\text{Fe}^{2+}/\text{Fe}}^0 = -0.44\text{V}$ and oxygen electrode $\text{O}_2 + 4\text{H}^+ + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}$; $E^0 = 1.23\text{V}$.

Ans: For oxygen electrode the variation of electrode potential with pH is given by

$$E = E^0 - \frac{0.0591}{n} \log Q$$

$$\text{Where } E_0 = 1.23\text{V} \text{ and } Q = \frac{1}{[\text{H}^+]^4}$$

$$E = 1.23 - \frac{0.0591}{4} \log \frac{1}{[\text{H}^+]^4}$$

$$E = 1.23 - 0.0591 \times \text{pH}$$

case 1: When $\text{pH} = 0$ ie: 1 M HCl; $E = 1.23 - 0.0591 \times 0 = 1.23\text{ V}$

case 2: When $\text{pH} = 14$ ie: Alkaline water; $E = 1.23 - 0.0591 \times 14 = 0.40\text{ V}$

$$E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}}$$

$$\text{when pH}=0 \text{ (1 M acid)} \quad E_{\text{cell}} = 1.23 - (-0.44) = 1.67\text{ V}$$

$$\text{when pH}=14 \text{ (1 M base)} \quad E_{\text{cell}} = 0.40 - (-0.44) = 0.84\text{ V}$$

When corrosion cell potential is more rate of corrosion will be more since free energy $\Delta G = -nFE$. Therefore, corrosion of iron will be more rapid in acid medium.

Problem 1.6.6: When an iron nail is immersed in 1 M HCl which of the following reactions will couple together? Following reduction potentials are given $E_{Fe^{3+}/Fe}^0 = -0.04\text{ V}$; $E_{Fe^{2+}/Fe}^0 = -0.44\text{ V}$; $E_{H^+/H_2}^0 = 0\text{ V}$; $E_{Cl_2/Cl^-}^0 = +1.36\text{ V}$.

Ans: Cell reaction having maximum emf is more spontaneous. Here H^+ undergoes reduction to H_2 gas since reduction potential is greater.

Fe can oxidize to Fe^{2+} or Fe^{3+}

In the formation of Fe^{3+} ; $E_{cell} = 0 - (-0.04) = +0.04\text{ V}$

In the formation of Fe^{2+} ; $E_{cell} = 0 - (-0.44) = +0.44\text{ V}$

Here Fe will oxidize to Fe^{2+} since its corrosion cell potential is more. Thus, evolution of H_2 couples with oxidation of Fe to Fe^{2+} . Here Cl^- cannot oxidize to Cl_2 as its reduction potential is very high.

Problem 1.6.7: What happens when a copper rod immersed in $FeCl_3$ solution. Justify the answer from following reduction potentials. $E_{Fe^{3+}/Fe}^0 = -0.04\text{ V}$; $E_{Fe^{3+}/Fe^{2+}}^0 = 0.77\text{ V}$; $E_{Cu^{2+}/Cu}^0 = +0.34\text{ V}$.

Ans: Here the reduction potential of $E_{Fe^{3+}/Fe^{2+}}^0 = 0.77\text{ V}$ is maximum hence reduction Fe^{3+} to Fe^{2+} takes place. $Fe^{3+} + e^- \rightarrow Fe^{2+}$ and copper metal will undergo oxidation $Cu \rightarrow Cu^{2+} + 2e^-$. Here no metal displacement reaction takes place.

Problem 1.6.8: When potassium permanganate solution is added drop wise to an acidified mixture of NaCl and KI predict the oxidation product. Given that $E_{Cl_2/Cl^-}^0 = +1.36\text{ V}$; $E_{I_2/I^-}^0 = 0.54\text{ V}$; $E_{MnO_4^-/H^+/Mn^{2+}}^0 = 1.51\text{ V}$.

Ans: More spontaneous reaction is reaction having maximum cell emf.

$$E_{cell} = E_{cathode} - E_{anode}$$

In the formation of Cl_2 ; $E_{cell} = 1.51 - 1.36 = +0.15\text{ V}$

In the formation of I_2 ; $E_{cell} = 1.51 - 0.54 = +0.97\text{ V}$

Here I^- from KI will oxidize to I_2 first. liberation of Cl_2 will happen only after all available KI get consumed

Problem 1.6.9: What are the products of electrolysis when we electrolyze an acidified mixture of NaCl and KI solution using platinum electrode. Given that $E_{Cl_2/Cl^-}^0 = +1.36\text{ V}$; $E_{I_2/I^-}^0 = 0.54\text{ V}$; $E_{H^+/H_2}^0 = 0\text{ V}$; $E_{Na^+/Na}^0 = -2.71\text{ V}$; $E_{K^+/K}^0 = -2.93\text{ V}$.

Ans: Cathode(-): Ions approaching cathode(-) are Na^+ , K^+ , H^+ . Here H^+ has highest reduction potential ($E_{H^+/H_2}^0 = 0\text{ V}$). Therefore, H^+ ion is reduced to H_2 gas. ($H^+ + e^- \rightarrow \frac{1}{2}H_2$)
Anode(+): Ions approaching anode(+) are Cl^- , I^- . Here I^- has lowest reduction potential ($E_{I_2/I^-}^0 = 0.54\text{ V}$). Hence, I^- get oxidized to I_2 ($2I^- \rightarrow I_2 + 2e^-$).

Here $E_{cell} = E_{cathode} - E_{anode} = 0 - 0.54 = -0.54$ V. In electrolytic cell E_{cell} is -ve. In electrolysis -ve value of E_{cell} should be a minimum.

Problem 1.6.10: What are the products of electrolysis of aqueous sodium chloride solution at cathode and anode using copper electrodes. Given that $E_{Cl_2/Cl^-}^0 = +1.36$ V; $E_{H^+/H_2}^0 = 0$ V at pH = 7; $E_{O_2/OH^-}^0 = +0.82$ V at pH = 7; $E_{Na^+/Na}^0 = -2.71$ V; and $E_{Cu^{2+}/Cu}^0 = +0.34$ V.

Ans: Cathode(-) Ions approaching cathode(-) are Na^+ and H^+ . Here H^+ has highest (less -ve) reduction potential. Therefore, H^+ ion is reduced to H_2 gas rather than reduction of Na^+ to Na metal.

Anode(+): Ions approaching anode(+) are Cl^- and OH^- , but Cu metal is also present at anode. Here Cu metal has lowest reduction potential, Hence Cu metal get oxidized to Cu^{2+} rather than the formation of O_2 or Cl_2 gas.

Problem 1.6.11: An electrochemical cell reaction involving 2 elections shows an emf of 1.3123 V at 30°C and 1.3178 V at 40°C. Find the standard emf of the cell and reaction quotient Q.

Ans:

$$E_{cell} = E_{cell}^0 - \frac{2.303RT}{2F} \log Q$$

When Q is a constant plot of E (E_{cell}) vs Temperature(T) must be a straight-line having slope B and intercept A. the equation has the form

$$E = A + BT$$

$$\text{At } 30^\circ\text{C}; 1.3123 = A + B \times 303 \quad (1)$$

$$\text{At } 40^\circ\text{C}; 1.3178 = A + B \times 313 \quad (2)$$

Equation (2)-(1) gives

$$\begin{aligned} 0.0055 &= B \times 10 \\ B &= 0.00055 \end{aligned}$$

substituting in (1)

$$\begin{aligned} 1.3123 &= A + 0.00055 \times 303 \\ A &= 1.14565V \quad \text{or} \quad E_{cell}^0 = 1.14565V \\ B &= -\frac{2.303R}{2F} \log Q \\ 0.00055 &= -\frac{2.303 \times 8.314}{2 \times 96500} \log Q \\ \log Q &= -5.5439 \\ Q &= 2.858 \times 10^{-6} \end{aligned}$$

1.7 Problems

1. Calculate the single electrode potential for Cu metal in contact with 0.15 M Cu^{2+} solution. $E_{\text{Cu}^{2+}/\text{Cu}}^0 = -0.34\text{V}$.
(ans: -0.315 V)
2. Calculate the emf of the cell $\text{Zn}/\text{Zn}^{2+}_{0.1\text{M}}//\text{Cu}^{2+}_{0.1\text{M}}/\text{Cu}$. Write its Half-cell and net cell reactions. $E_{\text{Zn}^{2+}/\text{Zn}}^0 = -0.76\text{V}$ and $E_{\text{Cu}^{2+}/\text{Cu}}^0 = 0.34\text{V}$.
(ans: 1.1V)
3. A Zn rod is placed in 0.1 M solution of ZnSO_4 at 25°C . Assuming that the salt is dissociated to the extent of 96 percentage at this dilution, Calculate the potential of the electrode? $E_{\text{Zn}^{2+}/\text{Zn}}^0 = -0.76\text{V}$. (ans: -0.79 V)
4. A Cu rod is placed in 0.1M solution of CuSO_4 at 50°C . Calculate the potential of the electrode? $E_{\text{Cu}^{2+}/\text{Cu}}^0 = +0.34\text{V}$. (ans: +0.307 V)
5. Write the cell reaction and cell representation of a Fe/Fe^{2+} coupled with a standard calomel electrode at 25°C . Given that $E_{\text{Fe}^{2+}/\text{Fe}}^0 = -0.44\text{V}$; $E_{\text{calomel}}^0 = 0.2810\text{V}$.
6. Potential of H_2 gas electrode set up in an acid solution of unknown strength is 0.0295 V at 25°C when coupled with SHE. Find the pH of the acid solution? (ans: 0.5)
7. A hydrogen electrode is immersed in a solution of pH = 3.3 and coupled with SHE. Calculate the emf. of the cell at 25°C ? (ans: 0.195 V)
8. (a) Derive Nernst equation for a cell reaction. Explain the applications of the equation.
(b) Calculate equilibrium constant for the reaction. $\text{Cu} + 2\text{Ag}^+ \rightarrow \text{Cu}^{2+} + 2\text{Ag}$. Given that $E^0 = 0.46\text{V}$. (ans: 3.68×10^{16})

1.8 Questions

1.8.1 Short answer questions

1. What is electrode potential? Explain Helmholtz double layer theory for its origin?
2. Formulate the electrochemical cell in which the reaction is $\text{Zn}_{(s)} + \text{Ag}^+ \rightarrow \text{Zn}^{2+} + 2\text{Ag}_{(s)}$ and write the half - cell reactions.
3. Distinguish between reversible and irreversible cells with examples.
4. What do you mean by a reference electrode, mention any two.
5. Why a minimum of two calibrations standards are required for calibration of pH meter?
6. Calculate the equilibrium constant of the following reaction at 25°C , $2\text{Li} + \text{Cu}^{2+} \rightarrow 2\text{Li}^+ + \text{Cu}$; Given that $E_{\text{Li}^+/\text{Li}}^0 = -3.05\text{V}$; $E_{\text{Cu}^{2+}/\text{Cu}}^0 = 0.34\text{V}$.
7. What is the origin of single electrode potential?

8. In redox titration using $KMnO_4$ as oxidizing agent, dil. HCl acid cannot be used to provide acidic medium, whereas dil. H_2SO_4 can be used, explain using following standard reduction potentials $E_{MnO_4^-/H^+}^0 = -1.51V$; $E_{S_2O_8^{2-}/SO_4^{2-}}^0 = +1.98V$; $E_{Cl_2/Cl^-}^0 = +1.36V$.
9. In redox titration using $K_2Cr_2O_7$ as oxidizing agent both dil. HCl acid as well as dil. H_2SO_4 can be used to provide acid medium, explain using standard reduction potentials $E_{Cr_2O_7^{2-}/H^+}^0 = -1.33V$; $E_{S_2O_8^{2-}/SO_4^{2-}}^0 = +1.98V$; $E_{Cl_2/Cl^-}^0 = +1.36V$.
10. Write the metal displacement reaction takes place when a Fe rod is dipped in $CuSO_4$ solution, justify based on standard reduction potentials $E_{Cu^{2+}/Cu}^0 = 0.34V$; $E_{Fe^{2+}/Fe}^0 = -0.44V$; $E_{Fe^{3+}/Fe^{2+}}^0 = -0.04V$.
11. Give the electrochemical mechanism of etching of copper using ferric chloride solution. Give the reaction and explain products formed. Given that $E_{Cu^{2+}/Cu}^0 = 0.34V$; $E_{Fe^{3+}/Fe^{2+}}^0 = -0.77V$.
12. Potassium dichromate solution is added to an acidified solution of a mixture of KI and $FeSO_4$, in drops, which will oxidize first, predict the reaction spontaneity using following standard reduction potentials $E_{Fe^{3+}/Fe^{2+}}^0 = -0.77V$; $E_{I_2/I^-}^0 = 0.54V$; $E_{Cr_2O_7^{2-}/H^+}^0 = +1.33V$.
13. Why Mg corrodes in both acidic and alkaline oxygen deficient conditions, whereas Fe does not corrode in alkaline oxygen deficient condition? $Mg^{2+} + 2e^- \rightarrow Mg$, $E^0 = -2.36V$, $Fe^{2+} + 2e^- \rightarrow Fe$, $E^0 = -0.44V$, $H^+ + e^- \rightarrow \frac{1}{2}H_2$, $E^0 = 0V$.
14. Which of the following metals can displace Copper from $CuSO_4$ solution (a) Fe (b) Ag (c) Zn (d) $Gold$. Justify your answer.
15. Calculate the conductivity of a given sample of water at $298K$ which shows a conductance of $620\mu S$ in the given cell at $298K$. A standard solution of $0.1M$ KCl shows a conductance of $12.34mS$ in that cell. (Given that conductivity of $0.1M$ KCl at $298K$ is $0.01288Scm^{-1}$).
16. Differentiate galvanic and electrochemical series.
17. Why lithium-ion cells are not allowed to charge completely?

1.8.2 Essay questions

1. Explain any four applications of Nernst equation.
2. Explain the applications of electrochemical series with suitable example.
3. What is glass electrode? Explain the construction and working of glass electrode used for pH measurement.
4. What is a calomel electrode? Explain its working and write the electrode reaction when it acts as an anode and a cathode.
5. Derive the Nernst equation for the cell EMF of Daniel cell.

6. Explain the construction and working of hydrogen-oxygen fuel cell which uses acid electrolyte in it.
7. Write a short note on the construction and working of Li-ion cell.
8. What are reference electrodes? Give examples for primary reference and secondary reference electrodes and give their electrode reactions.
9. What are the products of electrolysis of aqueous sodium chloride solution at cathode and anode using iron electrodes. Given that $E_{Cl_2/Cl^-}^0 = +1.36\text{ V}$; $E_{H^+/H_2}^0 = 0\text{ V}$; at $\text{pH} = 7$ $E_{O_2/OH^-}^0 = +0.82\text{ V}$; $E_{Na^+/Na}^0 = -2.71\text{ V}$; and $E_{Fe^{2+}/Fe}^0 = -0.44\text{ V}$; $E_{Fe^{3+}/Fe}^0 = -0.04\text{ V}$.
(Ans: Cathode (-) H_2 gas and anode (+) Fe^{2+})
10. What is a calomel electrode? How does the single electrode potential of calomel electrode vary with concentration of Cl^- ?
11. Give the relation between emf and pH of glass electrode calomel electrode assembly. Draw emf versus pH graph?
12. How will you determine the standard electrode potential of Fe/Fe^{2+} electrode using calomel electrode? Write the cell reaction and cell representation.
13. Define conductivity. How cell constant is determined? How conductivity of a water sample is determined?
14. Explain the mechanism of electrochemical corrosion of iron in different environmental conditions.
15. Explain various types of cathodic protection.
16. Explain the principle and procedure for electroless copper plating, give any two applications.
17. Explain the principle and procedure for copper electro plating, give any two applications.